Disentangling the Expressivity of RoPE

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Abstract

Two accounts recur in explanations of the success of rotary position embeddings (RoPE). Expressivity studies associate periodic position information with *modular predicates*, whereas mechanistic and long-context studies emphasize positional *anchors* and *local offsets*. We formalize both accounts for fully uniform, finite-precision soft-attention transformers. We find that, if every rotary component is *periodic*, RoPE transformers recognize exactly the languages definable in past temporal logic with modular predicates. Conventional RoPE is *different*: The rotations it computes never repeat. This yields a precision-dependent bounded simulation of fixed-offset *look-back* operators, rather than an all-length modular characterization. Controlled experiments match this separation: Constructed periodic schedules length-generalize on modular languages, while conventional RoPE behaves more like a bounded locality bias and can impair tasks requiring position-invariant access to distant context. Altogether, our findings shed light on RoPE transformers, bringing theoretical expressivity characterizations closer to models used in practice.

1 Introduction

In the absence of positional information, self-attention—the core operation of transformers (Vaswani et al., 2017)—is permutation-equivariant. Positional encodings (PEs) help distinguish symbol positions. The original transformer employs sinusoidal PEs (SiPE), which add position-dependent sine and cosine features to symbol representations. However, models using SiPE often generalize poorly to sequences longer than those seen during training (Rosendahl et al., 2019; Neishi & Yoshinaga, 2019; Press et al., 2022). Motivated by this limitation, RoPE was proposed as an alternative that encodes relative positions within attention: It rotates query and key vectors as a function of position, making their dot product depend on relative displacement (Su et al., 2023). RoPE has since become the de facto PE scheme in transformer-based LMs (Bai et al., 2023; Grattafiori et al., 2024; Team et al., 2024; Olmo et al., 2025; Yang et al., 2025a; DeepSeek-AI et al., 2026, *inter alia*).

Despite its widespread use, the theoretical contributions of RoPE remain unclear. Existing explanations fall broadly into two themes. Theoretical expressivity work links the periodicity of sinusoidal functions to modular predicates in formal logic (Yang et al., 2024; 2025b). More practically oriented analyses find that the apparent length generalization ability of RoPE can be superficial: RoPE may extrapolate to longer sequences without enabling effective use of distant context (Kazemnejad et al., 2023; Men et al., 2024; Gelberg et al., 2025; Du et al., 2026). A complementary, more mechanistic line of work instead emphasizes its role as a positional anchor and its bias toward fixed local offsets (Barbero et al., 2025a;b).

These perspectives *prima facie* appear incompatible. We reconcile them within the fully uniform fixed-precision transformer framework. We show that component-periodic RoPE (RoPE_P) is equivalent to past temporal logic with modular predicates, $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$. Conventional RoPE, in contrast, realizes a non-periodic schedule. The two variants differ only in the angles they rotate by. RoPE_P rotates the key and query subspaces by rational multiples

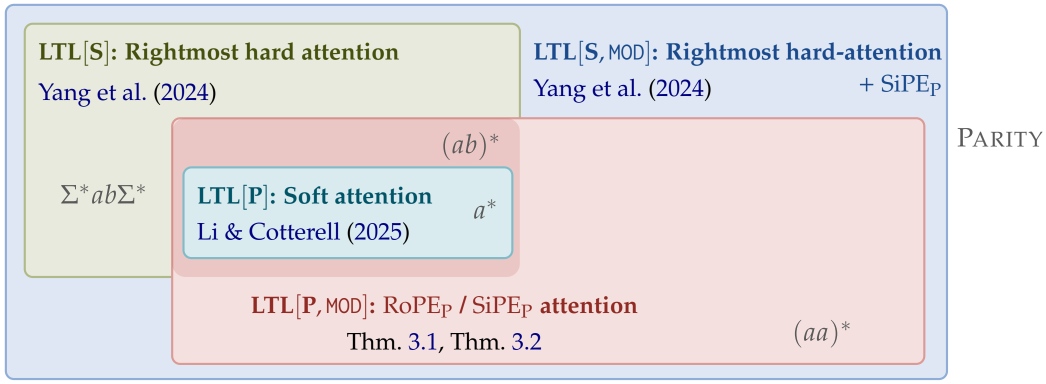

Figure 1: The expressivity landscape of fully uniform finite-precision transformers. The red box shows our contributions: The connection of RoPE$_P$ and SiPE$_P$ transformers to **LTL**[**P**, MOD], which is incomparable with **LTL**[**S**] (capturing rightmost hard-attention NoPE transformers). Their intersection contains **LTL**[**P**], corresponding to soft-attention NoPE transformers.

of 2π, so each rotary component cycles through finitely many phases and repeats with a fixed period, which a fixed-precision model can realize by a lookup table indexed by the position modulo that period. Conventional RoPE instead rotates by frequencies whose phases are incommensurate with 2π and therefore never repeat (§2.4). Its relative scores can simulate the look-back of k steps, only while a precision-dependent score gap is maintained. This connects RoPE to local attention (Li & Cotterell, 2026) and gives a formal account of fixed-offset heads observed in practice (Barbero et al., 2025a). Fig. 1 summarizes the landscape of fully uniform finite-precision transformers.

We vet our results experimentally by training transformers on controlled formal languages. We find that constructed periodic RoPE schedules length-generalize on languages requiring modular predicates. In contrast, no conventional base value generalizes on $(aa)^*$ or $(ab)^*$, suggesting that changing the base does not recover the modular construction. Across the same sweep, the mean longest-perfect length of conventional RoPE outperforms NoPE at six of seven bases on an **LTL**[**Y**] language, but underperforms at every base on both an **LTL**[**P**] language and a language requiring both **P** and **Y**. These results support the locality account while suggesting that the resulting local bias can interfere with long-distance conditioning.

2 Preliminaries

An **alphabet** Σ is a finite, non-empty set of **symbols** and Σ^* denotes the set of all strings over Σ. With $|\cdot|$, we denote the length of a string—in particular, we write $|w_1 \cdots w_N| = N$. We write ε for the **empty string**, the unique string of length 0. A **formal language** $\mathbb{L}$ over Σ is a subset of Σ^*.

2.1 Linear Temporal Logic

Regular languages are languages that can be described by regular expressions; see §A.3 for details. *Subclasses* of regular languages can equivalently be described with different formalisms such as automata and logic. Past work (Yang et al., 2024; Li & Cotterell, 2025; Jerad et al., 2025) links finite-precision transformers with subregular classes of languages. We use linear temporal logic (**LTL**) as our primary descriptive formalism.

Given $\boldsymbol{w} = w_1 \cdots w_N$, **LTL** formulas are evaluated at a position i; a formula accepts $\boldsymbol{w}$ when it holds at the readout position $N+1$. The atomic predicate $\pi_a(i)$ tests whether the symbol w_i at position i matches a. Furthermore, **LTL** leverages *temporal* operators to evaluate context-dependent properties at positions. The operators we need are the following:

$$\boldsymbol{w}, i \models \mathbf{P}\psi \iff \exists j < i : \boldsymbol{w}, j \models \psi, \qquad \boldsymbol{w}, i \models \mathbf{Y}\psi \iff i > 1 \wedge \boldsymbol{w}, i-1 \models \psi. \tag{1}$$

For $k \geq 1$, we write $\mathbf{Y}^k\psi$ for k-fold composition of $\mathbf{Y}$; equivalently, $\boldsymbol{w}, i \models \mathbf{Y}^k\psi$ if and only if $i > k$ and $\boldsymbol{w}, i-k \models \psi$. We write $\mathbf{LTL}[\mathbf{P}]$ and $\mathbf{LTL}[\mathbf{P},\mathbf{Y}]$ for the fragments generated from these operators, atomic predicates, and Boolean connectives. The detailed **LTL** semantics and equivalent first-order, automata, and algebraic formalisms appear in §A. A third operator, **S** ("since"), subsumes both **P** and **Y** (Gabbay et al., 1980), so $\mathbf{LTL}[\mathbf{S}]$ is all of **LTL**—exactly the star-free languages (Kamp, 1968; McNaughton & Papert, 1971). We use **first-order logic** over the linear order $<$ interchangeably with **LTL**, via $\mathbf{LTL}[\mathbf{S}] = \mathbf{FO}[<]$ (Kamp, 1968) and $\mathbf{LTL}[\mathbf{P}] = \mathbf{PFO}^2[<]$, the past fragment of the two-variable logic $\mathbf{FO}^2[<]$ (Li & Cotterell, 2025); §A.2 gives the formal definitions.

Modular predicates. The unary predicate $\texttt{MOD}^r_m(i)$ holds exactly when $i \equiv r \pmod{m}$. We write $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ for $\mathbf{LTL}[\mathbf{P}]$ augmented with all such predicates. Adding modular predicates preserves the **LTL–FO**$[<]$ correspondence: $\mathbf{LTL}[\mathbf{P},\texttt{MOD}] = \mathbf{PFO}^2[<,\texttt{MOD}]$ (Thm. 3.3).

2.2 Transformers

A consistent takeaway of theoretical work is that transformers' computational abilities heavily depend on the modeling assumptions one makes (Hao et al., 2022; Jerad et al., 2025; Svete et al., 2026). Understanding the *minimal viable* definition of a transformer is particularly useful, as it facilitates the analysis of the fundamental components of every transformer. To this end, we adopt the fully uniform fixed-precision model of Li & Cotterell (2025). Uniformity requires *all strings* be processed by a model defined by a single set of parameters, in contrast to having a separate model for each length.

Finite precision. Transformer parameters and activations belong to a *finite* set $\mathbb{F} \stackrel{\text{def}}{=} \{f_1, \ldots, f_K\}$, corresponding to real-world transformers (e.g., 32-bit floating-point arithmetic). All computations are performed rounded to $\mathbb{F}$, with ∞ and $-\infty$ in $\mathbb{F}$ which describe overflow. We denote by f_{large} the smallest positive element of $\mathbb{F}$ such that $\exp\left(-f_{\text{large}}\right)$ rounds to 0. This permits exact hard selection whenever the winning score exceeds every alternative by at least f_{large}.

Transformers as string classifiers. We study transformers as **string classifiers**. An input to a transformer is a string $\boldsymbol{w} \in \Sigma^*$ augmented with the beginning-of-string BOS and end-of-string EOS symbols, BOS$\boldsymbol{w}$EOS. We set $\overline{\Sigma} \stackrel{\text{def}}{=} \Sigma \cup \{\text{BOS}, \text{EOS}\}$. An initial embedding layer maps symbols in $\overline{\Sigma}$ to distinct elements of $\mathbb{F}^D$, where $D \in \mathbb{N}$ is the hidden state size. The embedding layer is followed by a stack of L transformer layers, each composed of an attention layer, a feedforward network, and layer-normalization. For a layer ℓ and position i, $\boldsymbol{x}^\ell_i \in \mathbb{F}^D$ is the **contextual representation** of symbol w_i after being processed by layer ℓ. We denote by $\boldsymbol{\lambda}\colon (\overline{\Sigma})^* \to (\mathbb{F}^D)^*$ the function that maps an input BOS$\boldsymbol{w}$EOS to contextual representations through the embedding layer and the L transformer layers. For a string $\boldsymbol{w}$ and string position i, we denote by $\boldsymbol{\lambda}(\boldsymbol{w})_i \in \mathbb{F}^D$ the final i'th contextual representation $\boldsymbol{x}^L_i$. To enable language recognition, $\boldsymbol{\lambda}$ is paired with an output classification layer $\mathcal{F}\colon (\mathbb{F}^D) \to \{0,1\}$ acting on $\boldsymbol{\lambda}(\boldsymbol{w})_{\text{EOS}}$ for an input $\boldsymbol{w}$. Given a precision regime $\mathbb{F}$, the tuple $\mathsf{T} \stackrel{\text{def}}{=} (\boldsymbol{\lambda}, \mathcal{F})$ is an $\mathbb{F}$-**transformer** and defines a language $\mathbb{L}(\mathsf{T}) \stackrel{\text{def}}{=} \{\boldsymbol{w} \mid \mathcal{F}(\boldsymbol{\lambda}(\boldsymbol{w})_{\text{EOS}}) = 1\}$.

Soft attention. We assume soft attention, which corresponds to standard implementations. We note a minute but important practical detail: For numerical stability, soft attention is usually computed by subtracting from $\boldsymbol{S}_{i,j}$—the attention score between the query $\boldsymbol{q}_i$ and the key $\boldsymbol{k}_j$—the maximum score. We define the normalized score $\boldsymbol{\alpha}_{i,j}$ as follows:

$$\boldsymbol{\alpha}_{i,j} = \frac{\exp\left(\boldsymbol{S}_{i,j} - \max_{k<i} \boldsymbol{S}_{i,k}\right)}{\sum_{l=0}^{i-1} \exp\left(\boldsymbol{S}_{i,l} - \max_{k<i} \boldsymbol{S}_{i,k}\right)}. \tag{2}$$

These NoPE transformers (**SMAT**) recognize exactly $\mathbf{LTL}[\mathbf{P}] = \mathbf{PFO}^2[<]$ (Li & Cotterell, 2025).

2.3 Rotary Position Embeddings

RoPE (Su et al., 2023) partitions the query's and key's D-dimensional space into two-dimensional subspaces and *rotates* each by an angle that depends on the absolute position i. Concretely, let

$$\boldsymbol{R}_{\Theta,i} \stackrel{\text{def}}{=} \operatorname{diag}\left(\mathbf{Rot}_{\theta_1,i}, \ldots, \mathbf{Rot}_{\theta_{D/2},i}\right), \qquad \mathbf{Rot}_{\theta,i} \stackrel{\text{def}}{=} \begin{pmatrix} \cos(\theta i) & -\sin(\theta i) \\ \sin(\theta i) & \cos(\theta i) \end{pmatrix} \tag{3}$$

where $\boldsymbol{R}_{\Theta,i}$ is a block-diagonal matrix and $\Theta \in [0, 2\pi]^{D/2}$ is a tuple of angles assigned to the different two-by-two rotation matrices in $\boldsymbol{R}_{\Theta,i}$, which we call the **schedule**. When the schedule is implicit, we drop it from the subscript and write, e.g., $\boldsymbol{R}_i$; we restore it whenever several schedules are in play. In conventional implementations, the angular frequencies are parameterized as $\theta_d = \beta^{-2(d-1)/D}$, where $\beta > 0$ is called the *base*; the original RoPE construction uses $\beta = 10^4$ (Su et al., 2023).

At position i, queries and keys become

$$\boldsymbol{q}_i = \boldsymbol{R}_{\Theta,i}\left(\boldsymbol{Q}\boldsymbol{x}_i + \boldsymbol{b}_q\right), \qquad \boldsymbol{k}_i = \boldsymbol{R}_{\Theta,i}\left(\boldsymbol{K}\boldsymbol{x}_i + \boldsymbol{b}_k\right). \tag{4}$$

For exact rotations and zero biases, orthogonality gives the relative-position identity

$$\boldsymbol{k}_j^\top \boldsymbol{q}_i = (\boldsymbol{R}_{\Theta,j}\boldsymbol{K}\boldsymbol{x}_j)^\top \boldsymbol{R}_{\Theta,i}\boldsymbol{Q}\boldsymbol{x}_i = \boldsymbol{x}_j^\top \boldsymbol{K}^\top (\boldsymbol{R}_{\Theta,j})^\top \boldsymbol{R}_{\Theta,i}\boldsymbol{Q}\boldsymbol{x}_i, = \boldsymbol{x}_j^\top \boldsymbol{K}^\top \boldsymbol{R}_{\Theta,i-j}\boldsymbol{Q}\boldsymbol{x}_i, \tag{5}$$

making the score depend on the distance $i - j$.

Ideal and realized rotations. The rotation above is **ideal**: It is evaluated in exact arithmetic, and it stores real numbers. A fixed-precision model computes something else. It stores a schedule $\widehat{\Theta}$ and evaluates the rotation with the finite-precision arithmetic. We write $\widehat{\boldsymbol{R}}_{\widehat{\Theta},i} \in \mathbb{F}^{D\times D}$ for the **realized** matrix in finite-precision arithmetic at position i, and $\widehat{\boldsymbol{R}}^{(d)}_{\widehat{\Theta},i} \in \mathbb{F}^{2\times 2}$ for its d^{th} block. The hat marks the realized map as opposed to the ideal one $i \mapsto \boldsymbol{R}_{\Theta,i}$. The distinction matters because rounding need not preserve the structure of the ideal map; our periodicity definitions concern the realized map.

Custom schedules and unrotated subspaces. To establish lower bounds for RoPE transformers, we permit a *custom* schedule—a bespoke set of angles Θ chosen for the target language. Custom schedules are standard in expressivity literature (e.g., Chiang et al., 2023; Yang et al., 2024; 2025b). We also allow *unrotated* subspaces with $\theta_d = 0$; these subsume partial-RoPE variants (Barbero et al., 2025b; Yang et al., 2025c; Khan et al., 2026) and retain the $\mathbf{LTL}[\mathbf{P}]$ computation available to NoPE, while separate rotary dimensions supply modular or fixed-offset information.

2.4 Component-Periodic and Non-Periodic RoPE

Intuitively, PEs with *periodicity* behave similarly to modular positional predicates in logic (Chiang et al., 2023; Yang & Chiang, 2024; Yang et al., 2025b). We formalize this intuition for finite-precision RoPE, with the proofs in §C.1. Throughout, $\widehat{\boldsymbol{R}}_i$ and $\widehat{\boldsymbol{R}}_i^{(d)}$ are the realized matrix and its d^{th} block.

Definition 2.1 (Component-periodic and non-periodic RoPE)**.** *A realized* RoPE *schedule* $\widehat{\Theta}$ *is* ***component-periodic*** *if, for every d, there is an integer $m_d \geq 1$ such that*

$$\widehat{\boldsymbol{R}}^{(d)}_{i+m_d} = \widehat{\boldsymbol{R}}^{(d)}_i \qquad \textit{for all } i \in \mathbb{N}. \tag{6}$$

It is ***non-periodic*** *if this condition fails: Some component has no positive integer period.*

Because D is fixed, component-periodicity is equivalent to periodicity of the full map $i \mapsto \widehat{\boldsymbol{R}}_i$. In particular, $M \stackrel{\text{def}}{=} \operatorname{lcm}(m_1, \ldots, m_{D/2})$ satisfies $\widehat{\boldsymbol{R}}_{i+M} = \widehat{\boldsymbol{R}}_i$ for every i, with unrotated blocks having period 1. We denote by **SMAT[**RoPE$_{\text{P}}$**]** and **SMAT[**RoPE$_{\text{NP}}$**]** the class of languages recognized by fixed-precision component-periodic and non-periodic RoPE transformers, respectively.

We show that periodic schedules can be expressed via finitely many modular predicates:

Proposition 2.1 (Component-periodicity via $\mathtt{MOD}$)**.** *If a realized* RoPE *schedule* $\widehat{\Theta}$ *is component-periodic, then its image is finite and, for every matrix coordinate and realized value, the positions attaining that value are definable by unary modular predicates.*

A standard class of periodic schedules is the set of **rational** multiples of π (Yang et al., 2024).

Proposition 2.2 (Rational-angle functions are periodic)**.** *If* $\theta_d/(2\pi) = a_d/m_d \in \mathbb{Q}$*, then* $i \mapsto \mathbf{Rot}_{\theta_d,i}$ *has period* m_d*.*

Intuitively, adding m_d to i changes the phase by $2\pi a_d$.

Prop. 2.2 concerns the *ideal* map, and it may not transfer to finite precision. The position i grows without bound as the string does, so neither i nor the phase $\theta_d i$ stays representable in $\mathbb{F}$ at all lengths. An implementation that forms the phase and then rounds therefore computes a map that need not agree with $\mathbf{Rot}_{\theta_d,i}$, and that realized map need not repeat. Rational angles alone thus do not make a schedule component-periodic in the sense of Def. 2.1. We resort to realizing the same rational-angle function without forming the product $\theta_d i$. For $\theta_d = 2\pi a_d/m_d$, we precompute the finite table

$$T_d(r) \stackrel{\text{def}}{=} \mathrm{round}_{\mathbb{F}}\left(\mathbf{Rot}_{\theta_d,r}\right), \qquad 0 \le r < m_d, \tag{7}$$

and define the realized block directly by $\widehat{\boldsymbol{R}}_i^{(d)} = T_d(i \bmod m_d)$. The table has m_d entries and the counter has m_d states, and both depend on the schedule but not on the input length, so the construction stays inside the fully uniform fixed-precision model. The realized map then satisfies $\widehat{\boldsymbol{R}}_{i+m_d}^{(d)} = \widehat{\boldsymbol{R}}_i^{(d)}$ for every i by construction. Similar cyclic lookups appear in practice. Huo (2026) precomputes one period of P-RoPE and indexes it by position modulo the period. Wang et al. (2024) round each RoPE wavelength to the nearest integer, so that every component repeats after a whole number of positions. Both were introduced to improve length generalization, and both are component-periodic in the sense of Def. 2.1, so the characterization we prove next applies to them directly.

3 A Characterization of SMAT[RoPE$_{\mathrm{P}}$]

This section describes the equivalence of **SMAT[**RoPE$_{\mathrm{P}}$**]** with $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}] = \mathbf{PFO}^2[<, \mathtt{MOD}]$. The next two subsections sketch the intuition for the lower and upper bounds; full proofs are in §C.

Theorem 3.1. **SMAT[**RoPE$_{\mathrm{P}}$**]** $= \mathbf{LTL}[\mathbf{P}, \mathtt{MOD}] = \mathbf{PFO}^2[<, \mathtt{MOD}]$.

3.1 From SMAT[RoPE$_{\mathrm{P}}$] to $\mathbf{PFO}^2[<, \mathtt{MOD}]$

Let M be the common period to all the rotation matrices in the block diagonal matrix (§2.4). Then, every realized entry depends only on $i \bmod M$, so equality to any realized value is a finite disjunction of predicates $\mathtt{MOD}_M^r(i)$ (Prop. 2.1). The fixed-precision simulation of all remaining transformer operations by $\mathbf{PFO}^2[<]$ (Li & Cotterell, 2025) then gives **SMAT[**RoPE$_{\mathrm{P}}$**]** $\subseteq \mathbf{PFO}^2[<, \mathtt{MOD}]$.

Lemma 3.1. *Any* $\mathbb{F}$*-transformer with* RoPE$_{\mathrm{P}}$ *can be simulated by* $\mathbf{PFO}^2[<, \mathtt{MOD}]$*. Hence,* **SMAT[**RoPE$_{\mathrm{P}}$**]** $\subseteq \mathbf{PFO}^2[<, \mathtt{MOD}] = \mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$.

RoPE **as a unary predicate.** Although RoPE scores encode *pairwise* distances, it does not require binary primitives; *unary* ones suffice. To see why, consider the binary modular predicate

$$\mathtt{MOD}_m^r(i,j) = \top \iff (j-i) \equiv r \pmod m, \tag{8}$$

which models the relative position dependency of RoPE by passing the distance between two positions through a periodic function. We show in §C.2 that the binary modular predicate can be expressed by a Boolean combination of unary modular predicates.

Proposition 3.1. *Binary modular predicates can be expressed with unary modular predicates and Boolean logic.*

3.2 From $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ to $\mathbf{SMAT}[\mathrm{RoPE}_{\mathrm{P}}]$

For the reverse inclusion, $\mathbb{F}$-transformers with $\mathrm{RoPE}_{\mathrm{P}}$ can first implement Boolean connectives and $\mathbf{P}$ in unrotated dimensions via structural induction (Li & Cotterell, 2025). For each modular atom $\mathtt{MOD}^{r}_{m}(i)$, a period-m rotary pair uses the lookup table $T_m(s) = \mathbf{Rot}_{2\pi/m,s}$ for $0 \leq s < m$ and returns $T_m(i \bmod m)$. An attention head compares the returned phase with the fixed BOS phase. A constant non-BOS baseline separates the matching residue from every nonmatching residue, and stable softmax turns the comparison into an exact Boolean value. Thus $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}] \subseteq \mathbf{SMAT}[\mathrm{RoPE}_{\mathrm{P}}]$.

Lemma 3.2. *For every* $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ *formula, there is an* $\mathbb{F}$*-transformer with* $\mathrm{RoPE}_{\mathrm{P}}$ *that simulates the formula. Hence* $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}] \subseteq \mathbf{SMAT}[\mathrm{RoPE}_{\mathrm{P}}]$.

Connection to attention sinks. The use of BOS as a fixed anchor connects the tightness construction to **attention sinks**, where language models allocate disproportionate attention to initial tokens (Xiao et al., 2024). One account of why sinks are useful is that they prevent contextual representations from *over-mixing* and keep them sufficiently distinct (Barbero et al., 2025b). Our construction offers a complementary reason: The anchor is what makes unary modular predicates computable, because the query's phase must be compared against a key whose own phase does not move. This suggests that sinks in RoPE transformers serve in part to expose positional phase, and we note that Gemma 7B, which uses RoPE, does concentrate attention on BOS (Barbero et al., 2025a).

3.3 Periodic Sinusoidal Positional Encodings

We find **absolute sinusoidal** PEs (Vaswani et al., 2017) to be equivalent to $\mathrm{RoPE}_{\mathrm{P}}$. As for $\mathrm{RoPE}_{\mathrm{P}}$, we denote by $\mathrm{SiPE}_{\mathrm{P}}$ the class of **periodic** sinusoidal encodings such that the realized sinusoidal position-to-embedding map has a finite integer period as in Def. 2.1. We similarly define $\mathrm{SiPE}_{\mathrm{NP}}$.

Theorem 3.2. $\mathbf{SMAT}[\mathrm{SiPE}_{\mathrm{P}}] = \mathbf{LTL}[\mathbf{P}, \mathtt{MOD}] = \mathbf{PFO}^2[<, \mathtt{MOD}]$.

The proof (§C.4) again expresses the finite periodic image with modular predicates and uses custom periodic features to recover each modular atom. This shows that the key factor for the corresponding all-length logical class is the *periodicity* of the encoding; both the absolute and relative encodings are unified under modular predicates.

3.4 Characterizing $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$

We have established that $\mathbf{SMAT}[\mathrm{RoPE}_{\mathrm{P}}]$ and $\mathbf{SMAT}[\mathrm{SiPE}_{\mathrm{P}}]$ can both be exactly characterized by $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$. To better grasp the expressive power of $\mathrm{RoPE}_{\mathrm{P}}$ and $\mathrm{SiPE}_{\mathrm{P}}$, we provide additional characterizations of $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$. Building on Dartois & Paperman (2013); Li & Cotterell (2025), we prove the following equivalences[1] in §B.

Theorem 3.3. *Let* $\mathbb{L} \subseteq \Sigma^*$ *be a regular language,* $\mathbb{M}$ *be its syntactic monoid, and* $\mathcal{A}$ *be the minimal DFA accepting it. The following assertions are equivalent:*

(i) $\mathbb{L}$ *is a left-deterministic modular polynomial*
(ii) $\mathbb{M} \in \mathbf{QR}$
(iii) $\exists k \in \mathbb{N}$ *such that* $\mathcal{A}_k$ *is partially ordered*
(iv) $\mathbb{L} = \mathbb{L}(\phi)$ *for some* $\mathbf{PFO}^2[<, \mathtt{MOD}]$ *formula* ϕ
(v) $\mathbb{L} = \mathbb{L}(\psi)$ *for some* $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ *formula* ψ

[1] The relevant formal language theory tools are properly introduced in §A.

Language	Defining property	Class	Account
$(aa)^*$	even length	$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$; not star-free	modular
$(ab)^*$	*ab* repeated	$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$; not $\mathbf{LTL}[\mathbf{P}]$	modular
$a\Sigma^*$	starts with *a*	$\mathbf{LTL}[\mathbf{P}]$	locality
$\Sigma^* a$	ends with *a*	$\mathbf{LTL}[\mathbf{Y}]$; not $\mathbf{LTL}[\mathbf{P}]$	locality
$\Sigma^* ab\Sigma^*$	contains *ab*	$\mathbf{LTL}[\mathbf{P},\mathbf{Y}]$; locally testable	locality

Table 1: The formal languages we train on, each over an alphabet Σ with $|\Sigma| > 1$. The last column records which account the language probes: the modular account of §3 or the locality account of §4. §E.1 gives the full definitions.

The characterization provides novel tools for (in)expressibility statements about $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$. We can first show via $\mathbf{PFO}^2[<,\texttt{MOD}]$ that $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ can leverage modular predicates to recognize languages with a *periodic behavior* where a substring is repeated indefinitely.

Proposition 3.2. *For every nonempty word $w_1 \cdots w_m$, the language $(w_1 \cdots w_m)^*$ belongs to* $\mathbf{PFO}^2[<,\texttt{MOD}]$.

Proof. It is defined by $\texttt{MOD}_m^0 \wedge \forall i \bigwedge_{0\le j<m} \left(\texttt{MOD}_m^j(i) \Rightarrow \pi_{w_j}(i)\right)$, which is in $\mathbf{PFO}^2[<,\texttt{MOD}]$. ∎

Importantly, Prop. 3.2 implies that languages such as $(ab)^*$ and $(aa)^*$ belong to $\mathbf{PFO}^2[<,\texttt{MOD}]$, leading to the following conclusions.

$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ and $\mathbf{LTL}[\mathbf{S}]$ are incomparable. In other words, neither class contains the other. In one direction, $(aa)^*$ is not star-free (Yang et al., 2024) and so lies outside $\mathbf{LTL}[\mathbf{S}]$, yet Prop. 3.2 places it in $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$. In the other direction, $\mathbf{LTL}[\mathbf{S}]$ can recognize the **locally testable** languages (Zalcstein, 1972)—those that require checking that a finite sequence of symbols occur at *consecutive* positions. Written in **LTL**, this needs the **Y** operator: $\Sigma^* ab\Sigma^*$ is defined by $\mathbf{P}(\pi_b \wedge \mathbf{Y}\pi_a)$. The only temporal operator of $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ is **P**, which cannot step to an adjacent position, and we show in §B.1 that $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ contains no such language.

$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$ cannot perform modular counting. Modular predicates count *positions*, not *symbols*. Adding them to $\mathbf{LTL}[\mathbf{P}]$ therefore does not let it decide whether the number of occurrences of a symbol is divisible by an integer. The canonical example is PARITY, the set of bitstrings with an even number of 1s. We prove in §B.1 that PARITY is not $\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$.

3.5 Testing the Modular Account

We train transformer classifiers on the languages in Tab. 1, which also records the account each language probes. Training strings have length at most 40; test strings have lengths 41–500. Every configuration is run over 5 seeds and 3 learning rates, and every conventional-schedule configuration is fully rotated. Besides accuracy, we report the longest perfect length N^*: the largest N for which accuracy is 100% at every tested length through N. §E gives the architecture and optimization settings and reports per-run means and standard deviations. Tab. 2 compares NoPE with the period-targeting variants, while Fig. 2 reports the conventional non-periodic controls. In both variants, RoPE is applied on all dimensions.

		NoPE		RoPE$_\text{P}$		SiPE$_\text{P}$	
Language	$\in$ Class	Acc.	N^*	Acc.	N^*	Acc.	N^*
$(aa)^*$	$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$	0.50	41	**1.00**	**500**	0.50	41
$(ab)^*$	$\mathbf{LTL}[\mathbf{P},\texttt{MOD}]$	0.96	96	**1.00**	**500**	0.50	44

Table 2: Periodic-construction experiments. Maximum accuracy and longest perfect length over the run grid achieved by different transformer variants.

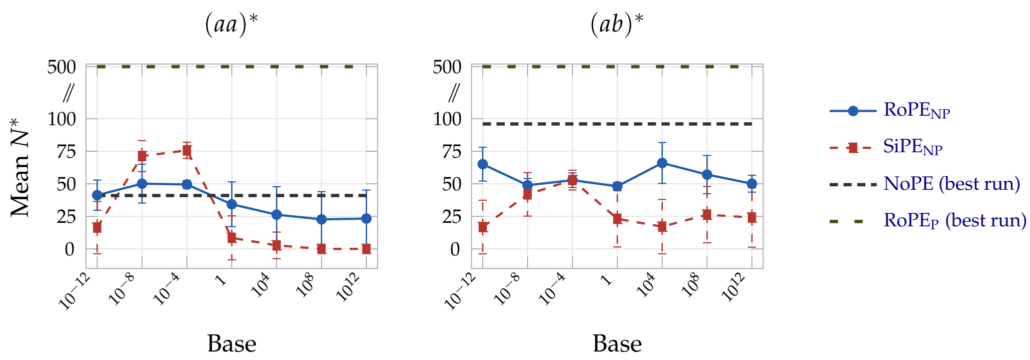

Figure 2: Mean longest perfect length N^* for conventional non-periodic encodings on the modular languages. Markers are means over the run grid and whiskers are ± 1 standard deviation. The two reference lines are best-run values for NoPE and RoPE$_{\text{P}}$ on these tasks. Neither non-periodic encoding generalizes perfectly at any tested base.

RoPE$_{\text{P}}$ **generalizes perfectly on LTL$[\mathbf{P}, \texttt{MOD}]$.** The period-targeting RoPE$_{\text{P}}$ model achieves accuracy 1.00 and $N^* = 500$ on both $(ab)^*$ and $(aa)^*$, matching the modular-predicate construction.

SiPE$_{\text{P}}$ **struggles on LTL$[\mathbf{P}, \texttt{MOD}]$.** Although SiPE$_{\text{P}}$ has the same existential expressivity, the trained model does not learn either modular language. The theorem guarantees a parameterization, not that optimization will find it; one possible explanation is that absolute sinusoidal features in the residual stream are more easily overwritten.

Conventional rotations do not learn the constructed modular behavior. Neither RoPE$_{\text{NP}}$ nor SiPE$_{\text{NP}}$ generalizes perfectly on the two periodic languages. The full sweep in Fig. 2 reaches the same conclusion at every tested base across twenty-four orders of magnitude, separating the engineered periodic behavior from conventional frequencies.

4 A Practical Account of Non-Periodic RoPE

We now analyze RoPE's role implementing fixed-offset heads (Barbero et al., 2025a)—referencing positions at a specific displacement from the query position. This account applies naturally to *conventional* RoPE$_{\text{NP}}$, but does not capture its *asymptotic* behavior. Rather, it relies on a finite-precision range over which RoPE can create a reliable fixed-offset head, partly explaining its empirical advantage over SiPE$_{\text{NP}}$ (Su et al., 2023). The proof for the bounded non-periodic RoPE simulations is in §D.

4.1 Conventional RoPE Is Non-Periodic

We first pinpoint why conventional RoPE is non-periodic. Namely, *irrational* rotations, those that conventional RoPE leverages, imply non-periodicity.

Proposition 4.1 (Specialization of Walters 1982, Thm 1.8)**.** *If $g/(2\pi) \notin \mathbb{Q}$, then $i \mapsto \mathrm{round}_{\mathbb{F}}\left(\mathbf{Rot}_{g,i}\right)$ is non-periodic. Hence, conventional* RoPE*, which contains $g = 1$, is non-periodic under output-rounded ideal-phase evaluation.*

4.2 Fixed-Offset Attention up to a Length Bound

We now proceed to show that with irrational rotations, RoPE$_{\text{NP}}$ transformers can implement fixed-offset attention heads.

Fix an offset $k \geq 1$. With exact arithmetic, a frequency $g/(2\pi) \notin \mathbb{Q}$ yields a k-offset head whose score is

$$S_{i,j}^{g,C,k} = C\cos\left(g\left(i - k - j\right)\right), \qquad 0 \leq j < i, \tag{9}$$

for some $C > 0$ (Barbero et al., 2025a). At positions $i \geq k+1$, the target $j = i - k$ uniquely maximizes Eq. (9) with score C. For the uniqueness to hold under finite precision, the target must remain ahead by at least f_{large} after rounding in the score computation.

We define the k-offset head's **certified range** as the maximum length at which it scores its target position at least f_{large} more than any non-target position:

$$N^{(k)}_{\max}\left(g, C, \mathbb{F}\right) \overset{\text{def}}{=} \sup\left\{L \in \mathbb{N} : \widehat{S}^{g,C,k,\mathbb{F}}_{i,i-k} - \widehat{S}^{g,C,k,\mathbb{F}}_{i,j} \geq f_{\text{large}} \ \substack{\text{for all } k+1 \leq i \leq L+1, \\ 0 \leq j < i,\ j \neq i-k}\right\}. \tag{10}$$

Under exact evaluation of Eq. (9) through length $L \geq k$, the sufficient gap is $C\delta_L\left(g\right)$, where $\delta_L\left(g\right) \overset{\text{def}}{=} 1 - \max_{1 \leq n \leq L} \cos\left(gn\right)$. Thus $C\delta_L\left(g\right) \geq f_{\text{large}}$ certifies $N^{(k)}_{\max}\left(g, C, \mathbb{F}\right) \geq L$.

4.3 From Fixed-Offset Heads to $\mathbf{Y}^k$

If the certified range is L, stable softmax makes position $i - k$'s weight exactly 1 for every query position $i \in \{k+1, \ldots L+1\}$. Unrotated dimensions independently simulate $\mathbf{LTL}[\mathbf{P}]$ and provide the boundary test for the first k positions.

Corollary 4.1. *Fix g, C, and $\mathbb{F}$, and let $L \in \mathbb{N}$. Write* **SMAT[**RoPE_{NP}**]**-*transformers for soft-attention* RoPE_{NP} *transformers with unrotated subspaces, and let L-simulation be as in Def. D.1.*

(i) For every offset $k \geq 1$ with $k \leq L \leq N^{(k)}_{\max}\left(g, C, \mathbb{F}\right)$, such a transformer L-simulates the fixed-offset operator $\mathbf{Y}^k$.

(ii) If $1 \leq L \leq N^{(1)}_{\max}\left(g, C, \mathbb{F}\right)$, it L-simulates every $\mathbf{LTL}[\mathbf{P}, \mathbf{Y}]$-definable formula. More generally, for a finite set of offsets $K \subseteq \mathbb{N}_{\geq 1}$ with $\max K \leq L \leq \min_{k \in K} N^{(k)}_{\max}\left(g, C, \mathbb{F}\right)$, it L-simulates every formula built from atomic predicates, Boolean connectives, $\mathbf{P}$, and the operators $\{\mathbf{Y}^k : k \in K\}$.

(iii) Moreover, for every $L \in \mathbb{N}$ and every finite offset set K there exist a rotation angle g with $g/\left(2\pi\right) \notin \mathbb{Q}$, scales C, and a finite-precision regime $\mathbb{F}$ satisfying the hypothesis of (ii).

The connection to $\mathbf{LTL}[\mathbf{P}, \mathbf{Y}]$ links the result to the extra power of local attention (Li & Cotterell, 2026) and resembles the finite-precision implementation of n-gram heads (Svete & Cotterell, 2024; Svete et al., 2024).

4.4 Testing the Locality Account

We evaluate whether the bounded fixed-offset mechanism appears in length generalization, under the protocol of §3.5. RoPE is again applied on all dimensions.

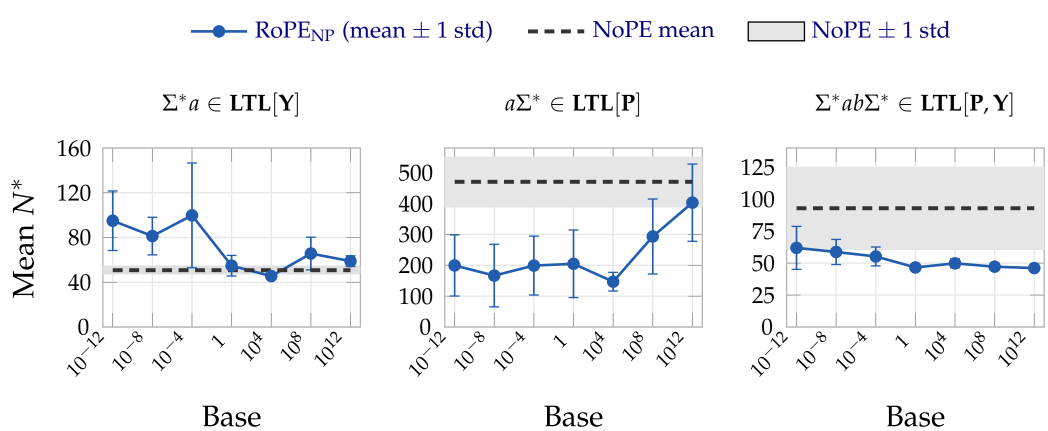

Figure 3: Mean longest perfect length N^*. Whiskers are ±1 standard deviation over the run grid and the grey band is the corresponding NoPE interval.

The standard base is not uniformly best. On each single-operator language, the conventional base $\beta = 10^4$ gives the lowest mean N^* among the seven tested bases: 45.5 on $\Sigma^* a$ and 146.9 on $a\Sigma^*$. Every tested base on either side of 10^4 improves on these two values. This mirrors the extrapolation experiments of Liu et al. (2024), who find 10^4 to be the worst tested base after fine-tuning and report improvements from both smaller and larger bases. The pattern is not universal, as, on the combined **P–Y** language, $10^1 2$ performs worse. This suggests that the optimal base is task-dependent.

RoPE_{NP} **improves on a Y language.** On $\Sigma^* a$, RoPE_{NP}'s mean N^* exceeds NoPE's at six of seven bases (Fig. 3). The certified-range result offers a mechanism: Relative scores can isolate the preceding position while their finite-precision gap remains large enough. As length grows, more competing offsets appear, increasing the chance of a numerical collision and making the behavior brittle.

RoPE_{NP} **can disrupt P reasoning.** On $a\Sigma^*$, the NoPE mean is 470.5, whereas the RoPE_{NP} means range from 146.9 to 402.9. On $\Sigma^* ab \Sigma^*$, the NoPE mean is 92.9, whereas the RoPE_{NP} means range from 46.1 to 61.9. This suggests a tradeoff: Relative displacement can help retrieve a nearby symbol while disrupting the nearly position-invariant aggregation that supports conditioning on distant information. Conventional RoPE therefore behaves more like a bounded source of **Y** access than a monotonic expressivity improvement over NoPE. We conjecture that partial-RoPE (Barbero et al., 2025a; Khan et al., 2026) and NoPE-RoPE hybrid attention (Yang et al., 2025c) variants should empirically improve on long-context **P** reasoning while retaining the locality bias of RoPE_{NP}.

5 Discussion

Two regimes. For component-periodic schedules, RoPE_{P} yields precisely $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ over arbitrary input lengths, enabling modular reasoning over input positions. The conventional RoPE_{NP} behaves qualitatively differently: Relative scores can directly identify any fixed offset k, but only while finite precision preserves a sufficient score gap, yielding bounded $\mathbf{Y}^k$ operators and a bounded simulation of $\mathbf{LTL}[\mathbf{P}, \mathbf{Y}]$. The two accounts therefore differ both in their schedule assumption and in whether the claim is uniform over all lengths. With unrotated subspaces, RoPE also retains the **P** reasoning of NoPE—checking whether a symbol occurs anywhere in the context.

Limitations. Our periodic lower bound uses a repeated finite table and a tailored exact value set, and does not imply that standard frequencies or optimization discover modular behavior. Empirical results, however, suggest that strong modular behavior often appears. The locality result of §4 is bounded and is not an all-length characterization of RoPE_{NP}.

Architectural implications. The comparison suggests reserving unrotated dimensions when a model must combine global **P** reasoning with periodic or local relative-position information. It also supports treating rotation structure as an explicit architectural choice, consistent with work questioning broad long-context claims for RoPE (Du et al., 2026). Neither account gives RoPE general state-tracking or regular language recognition abilities (Liu et al., 2023b), such as those required by Flip-Flop language modeling (Liu et al., 2023a). Alternative state-tracking mechanisms include rational transductors (Mohri, 2026) and transformer–linear-RNN hybrids (Merrill et al., 2026b;a). A remaining theoretical question is how the known depth hierarchy for RoPE transformers (Yang et al., 2025b) differs between exact periodic and bounded-local regimes.

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Appendix Contents

A Preliminaries: Formal Language Theory

We introduced formal languages and linear temporal logic in §2. We now provide the detailed **LTL** semantics, and introduce the tools that provide a richer understanding of fragments of **LTL**. We summarize in Tab. 3 known characterizations of relevant fragments of **LTL**. In §B, we give a proof of the exact characterization of **LTL**$[\mathbf{P}, \texttt{MOD}]$ in terms of polynomials, first-order logic, automata and the syntactic monoid. Our proofs are based on similar characterizations of **LTL**$[\mathbf{P}]$ and **LTL**$[\mathbf{P}, \mathbf{F}, \texttt{MOD}]$ (Li & Cotterell, 2025; Dartois & Paperman, 2013).

A.1 Linear Temporal Logic

The full semantics for **LTL** are the following.

- $\boldsymbol{w}, i \models \pi_a \iff w_i = a$;
- $\boldsymbol{w}, i \models \psi_1 \vee \psi_2 \iff \boldsymbol{w}, i \models \psi_1 \vee \boldsymbol{w}, i \models \psi_2$;
- $\boldsymbol{w}, i \models \psi_1 \wedge \psi_2 \iff \boldsymbol{w}, i \models \psi_1 \wedge \boldsymbol{w}, i \models \psi_2$;
- $\boldsymbol{w}, i \models \neg\psi \iff \boldsymbol{w}, i \not\models \psi$;
- $\boldsymbol{w}, i \models \mathbf{P}\psi \iff \exists j < i : \boldsymbol{w}, j \models \psi$;
- $\boldsymbol{w}, i \models \mathbf{F}\psi \iff \exists j > i : \boldsymbol{w}, j \models \psi$;
- $\boldsymbol{w}, i \models \psi_1 \mathbf{S} \psi_2 \iff \exists j < i : \boldsymbol{w}, j \models \psi_2$ and $\boldsymbol{w}, k \models \psi_1$ for all k with $j < k < i$;
- $\boldsymbol{w}, i \models \psi_1 \mathbf{U} \psi_2 \iff \exists j > i : \boldsymbol{w}, j \models \psi_2$ and $\boldsymbol{w}, k \models \psi_1$ for all k with $i < k < j$;
- $\boldsymbol{w}, i \models \mathbf{Y}\psi \iff \boldsymbol{w}, i - 1 \models \psi$ for $i > 1$;

To define string acceptance, we denote by $N+1$ a position outside of the string and define

$$\boldsymbol{w} \models \psi \Leftrightarrow \boldsymbol{w}, N+1 \models \psi. \tag{11}$$

For a set of temporal operators $\mathcal{O}$, we denote by $\mathbf{LTL}[\mathcal{O}]$ the corresponding fragment of **LTL**. We say a formula ψ is a $\mathbf{LTL}[\mathcal{O}]$ formula if it can be written in **LTL** with only the Boolean connectives and operators from $\mathcal{O}$. **S** subsumes the left-context-dependent operators **P** and **Y** (Gabbay et al., 1980).

Example:

The language $a\Sigma^*$ of strings starting with a specific symbol a can be described by a $\mathbf{LTL}[\mathbf{P}]$ formula:

$$\mathbf{P}\left(\pi_a \wedge \neg \mathbf{P}\top\right) \tag{12}$$

The language $\Sigma^* a$ of strings ending with a specific symbol a can be described by a $\mathbf{LTL}[\mathbf{Y}]$ formula:

$$\mathbf{Y}\pi_a \tag{13}$$

The language $a\Sigma^* a$ can be described by a $\mathbf{LTL}[\mathbf{S}]$ definable formula because **S** can express both **P** and **Y**.

LTL has become a key tool in characterizing the expressivity of fixed-precision transformers. Softmax transformers, leftmost hard-attention transformers and average hard-attention transformers are equivalent to $\mathbf{LTL}[\mathbf{P}]$ (Li & Cotterell, 2025; Jerad et al., 2025), while rightmost hard-attention transformers are equivalent to full $\mathbf{LTL}[\mathbf{S}]$ (Yang et al., 2024).

As formalized in §2, **LTL** can be extended with modular predicates, denoted by $\mathbf{LTL}[\mathcal{O}, \texttt{MOD}]$ for a set of operators $\mathcal{O}$.

A.2 First-Order Logic

First-order logic ($\mathbf{FO}[<]$) can also be used to characterize subregular classes. $\mathbf{FO}[<]$'s building blocks are **atomic** formulas, denoted by the unary predicate π_a for all $a \in \Sigma$ and the binary predicate $<$ which determines the order between string positions. We have $\pi_a(i) = \top$ if and only if $w_i = a$. Building on these atomic formulas, $\mathbf{FO}[<]$ operates on **variables**, which denote positions of the input string. In $\mathbf{FO}[<]$, a variable is either **free** or **bounded** by a quantifier. For example, in the existential quantification $\exists i < j$, i is a bounded variable while j is a free variable. $\mathbf{FO}[<]$ formulas are then constructed inductively as follows.

(1) Every atomic formula is an $\mathbf{FO}[<]$ formula;
(2) A Boolean combination of $\mathbf{FO}[<]$ formulas is an $\mathbf{FO}[<]$ formula;
(3) If $\phi(i\ldots)$ is an $\mathbf{FO}[<]$ formula, then so is $\exists i\colon \phi(i\ldots)$[2].

Let ϕ be a formula without any free variables. We say $\boldsymbol{w} \models \phi$ if ϕ is satisfied when evaluating it on $\boldsymbol{w}$. ϕ induces a language, which we denote by $\mathbb{L}(\phi) \stackrel{\text{def}}{=} \{\boldsymbol{w} \in \Sigma^* \mid \boldsymbol{w} \models \phi\}$.

Fragments of $\mathbf{FO}[<]$: We denote by $\mathbf{FO}^2[<]$ the fragment of $\mathbf{FO}[<]$ restricted to using at most two distinct variables in any sub-formula. We denote by $\mathbf{PFO}^2[<]$ the past fragment of $\mathbf{FO}^2[<]$ in which formulas are restricted so that whenever a free variable i is present, every quantifier must be of the form $\exists j\colon j < i$ or $\forall j\colon j < i$. In other words, only past positions relative to the free variable are accessed. $\mathbf{PFO}^2[<]$ formulas can be inductively written as follows.

(1) Every atomic formula is a $\mathbf{PFO}^2[<]$ formula;
(2) A Boolean combination of $\mathbf{PFO}^2[<]$ formulas is a $\mathbf{PFO}^2[<]$ formula if it does not contain more than two variables;
(3) If $\phi(i,j)$ is a $\mathbf{PFO}^2[<]$ formula with two free variables i, j, $\exists i < j\colon \phi(i,j)$ and $\exists j < i\colon \phi(i,j)$ are $\mathbf{PFO}^2[<]$ formulas;

[2] The universal quantifier $\forall$ is constructed by applying negation to the existential quantifier.

(4) If $\phi(i)$ is a $\mathbf{PFO}^2[<]$ formula with one free variable i, $\exists i < j \colon \phi(i)$ is a $\mathbf{PFO}^2[<]$ formula;
(5) If $\phi(i)$ is a $\mathbf{PFO}^2[<]$ formula with one free variable i, $\exists i \colon \phi(i)$ is a $\mathbf{PFO}^2[<]$ formula.

Over finite strings, $\mathbf{PFO}^2[<]$ and $\mathbf{LTL}[\mathbf{P}]$ define the same languages (Li & Cotterell, 2025), while $\mathbf{FO}[<]$ is equivalent to the full star-free class, which is $\mathbf{LTL}[\mathbf{S}]$ (Kamp, 1968; McNaughton & Papert, 1971; Gabbay et al., 1980). We therefore use $\mathbf{LTL}[\mathbf{P}]$ in the main narrative and retain $\mathbf{PFO}^2[<]$ when invoking first-order, automata-theoretic, or algebraic characterizations.

Modular predicates. Akin to **LTL**, $\mathbf{FO}[<]$ can be extended with unary modular predicates, albeit with a slight modification to its **LTL** counterpart. In **LTL**, formulas are evaluated at position $N+1$, meaning **LTL** can robustly make statements about the residue class of N, the length of the string. Because $\mathbf{FO}[<]$ formulas are evaluated by ranging variables over the entire string, such an operator cannot be directly implemented with unary predicates. Therefore, 0-ary predicates $\mathtt{MOD}_m^r$ are needed, where $\mathtt{MOD}_m^r$ is true if and only if the length of the input string N is congruent to r modulo m.

Adding these modular predicates to first-order fragments gives $\mathbf{PFO}^2[<, \mathtt{MOD}]$, $\mathbf{FO}^2[<, \mathtt{MOD}]$, and $\mathbf{FO}[<, \mathtt{MOD}]$; as shown in §3.4, $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ and $\mathbf{PFO}^2[<, \mathtt{MOD}]$ define the same class of languages.

A.3 Regular Expressions

Regular expressions are a declarative formalism for describing languages, defined recursively: (1) ε and each $w \in \Sigma$ is a regular expression; (2) If α and β are regular expressions, so are the union $\alpha + \beta$, concatenation $\alpha\beta$, complement α^C and the Kleene closure α^*. A language is **regular** if it can be described by a regular expression. A regular language is **star-free** if it can be described by a regular expression without the Kleene operator.

Definition A.1. *A **monomial** over an alphabet Σ is a language described via a regular expression of the form:*

$$\Sigma_0^* w_1 \Sigma_1^* \dots w_N \Sigma_N^* \tag{14}$$

Where $\Sigma_0, \Sigma_1 \dots \subseteq \Sigma$ and $w_1, w_2 \dots \in \Sigma$.

Definition A.2. *A **left-deterministic** monomial is a monomial where for every $0 < i \leq N$, $w_i \notin \Sigma_{i-1}$. A **right-deterministic** monomial is a monomial where for every $0 < i \leq N$, $w_i \notin \Sigma_i$. An **unambiguous** monomial is a monomial that is either left-deterministic or right-deterministic.*

Example:

The language $a\Sigma^*$ of strings starting with the symbol a is a left-deterministic monomial, since the marker a does not appear in the preceding empty block ($a \notin \Sigma_0$).

Symmetrically, the language $\Sigma^* a$ of strings ending with the symbol a is a right-deterministic monomial. Both $a\Sigma^*$ and $\Sigma^* a$ are therefore unambiguous monomials.

Their intersection $a\Sigma^* \cap \Sigma^* a = a\Sigma^* a$ is an unambiguous monomial.

A **polynomial** is a finite union of monomials. A left-deterministic (resp., right-deterministic, unambiguous) polynomial is a finite union of left-deterministic (resp., right-deterministic, unambiguous) monomials.

To adapt our analysis to RoPE transformers, we now introduce *modular* polynomials (Dartois & Paperman, 2013).

Definition A.3. *A monomial over Σ is **modular** with respect to some integer d if it can be written as:*

$$(\Sigma_{0,0}\Sigma_{0,1} \dots \Sigma_{0,d-1})^* w_1 (\Sigma_{1,0}\Sigma_{1,1} \dots \Sigma_{1,d-1})^* \dots w_N (\Sigma_{N,0}\Sigma_{N,1} \dots \Sigma_{N,d-1})^* \tag{15}$$

Where:

- *Each marker w_i occurs at a position congruent to some r_i modulo d;*

- *Each $\Sigma_{i,j}$ is some subset of Σ;*
- *For every i and every position p lying in the i'th block, the symbol at position p belongs to $\Sigma_{i,p \bmod d}$*

Note that the standard monomial is recovered with $d = 1$.

Definition A.4. *A* ***left-deterministic modular*** *monomial is a modular monomial such that for every $0 < i \leq N$, $w_i \notin \Sigma_{i-1,r_i}$.*

As before, we similarly define unambiguous modular monomials, unambiguous modular polynomials and left-deterministic modular polynomials.

A.4 Automata

The standard formalism for (sub)regular languages are finite automata—machines that transition within finitely many states.

Definition A.5. *A* ***semiautomaton*** *$\mathcal{A}$ is a 3-tuple (Σ, Q, δ) where Σ is an alphabet, Q is a finite set of* ***states*** *and $\delta\colon Q \times \Sigma \to Q$ is a* ***transition function****. We further define an* ***initialized semiautomaton*** *as a semiautomaton with an initial state.*

Definition A.6. *A* ***deterministic finite automaton (DFA)*** *$\mathcal{A}$ is a 5-tuple $(\Sigma, Q, q_\iota, F, \delta)$ where (Σ, Q, δ) is a semiautomaton, $q_\iota \in Q$ is an initial state, and $F \subseteq Q$ is a set of final states.*

We now introduce the automata that characterize the class of languages described by $\mathbf{PFO}^2[<]$. Intuitively, because quantifiers in $\mathbf{PFO}^2[<]$ can only look *backward*, they can only accumulate information about the past. We now introduce a class of automata that mirror this behavior: Their states are partially ordered and transitions may only move upward in this order. Consequently, reading a symbol can only refine what the automaton knows about the prefix read so far—no information can be discarded, and no state can be revisited.

Definition A.7. *Let $\delta^*\colon Q \times \Sigma^* \to Q$ be the* ***transitive closure*** *of δ, defined as*

$$\delta^*(q, w) = \delta(q, w), \text{ for } w \in \Sigma \tag{16a}$$

$$\delta^*(q, w_1 \cdots w_N) = \delta(\delta^*(q, w_1 \cdots w_{N-1}), w_N) \tag{16b}$$

with $\delta^*(q, \varepsilon) = q$ for any $q \in Q$. A ***partially ordered DFA (*PODFA*)*** *is a DFA $\mathcal{A} = (\Sigma, Q, q_\iota, F, \delta)$ where there is a partial order relation $\preceq$ on Q defined as $q \preceq p$ if and only if $\delta^*(q, \boldsymbol{w}) = p$ for some string $\boldsymbol{w} \in \Sigma^*$.*

As an example, the language of strings that contain the symbol a over any alphabet can be modeled by the following PODFA.

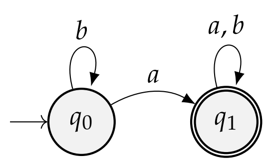

Figure 4: PODFA for $\Sigma^* a \Sigma^*$ over $\Sigma = \{a, b\}$. The partial order $q_0 \preceq q_1$ holds since q_1 is reachable from q_0 but not vice versa.

To link automata with modular predicates, we now introduce the k-automaton of some given automaton. In first-order logic with modular predicates, strings can be evaluated to the same value *up to a modulus*. For some modulus k, we therefore consider k-automata where strings are read as blocks of k symbols.

Definition A.8. *Let $\mathcal{A} = (\Sigma, Q, q_\iota, F, \delta)$ be a DFA. Let k be an integer. We define $\mathcal{A}_k \stackrel{\text{def}}{=} (\Sigma^k, Q, q_\iota, F, \delta^k)$ as the* ***k-automaton*** *of $\mathcal{A}$ where:*

- $\Sigma^k \stackrel{\text{def}}{=} \{\boldsymbol{w} \mid |\boldsymbol{w}| = k \text{ and } \boldsymbol{w} \in \Sigma^*\}$

- $\delta^k(q, w_1 w_2 \ldots w_k) \stackrel{\text{def}}{=} \delta(\ldots\delta(\delta(q, w_1), w_2), w_k)$.

We denote by PODFA_k the set of automata such that their k-automaton are PODFAs. As an example, $(ab)^*$ is the language of an automaton $\mathcal{A}$ in PODFA_2, as seen in Fig. 5.

Figure 5: Minimal automata for $(ab)^*$ and its corresponding 2-automaton.

A.5 The Syntactic Monoid

The **syntactic monoid** is a tool that partitions Σ^* into classes of strings, where each class contains strings that all contain the same *syntactic* information with respect to a given regular language. We now formalize the syntactic monoid of a language.

Definition A.9. *A* ***monoid*** $\mathbb{M}$ *is a set equipped with a binary operation and an identity element.*

For instance, the **free monoid** is the set Σ^* equipped with the concatenation operation and the empty string ε as identity.

Definition A.10. *A monoid* $\mathbb{M}$ *is in* **R** *if and only if for all* $w_1, w_2, w_3 \in \mathbb{M}$, $w_1 w_2 w_3 = w_1$ *implies* $w_1 w_2 = w_1$. *A monoid* $\mathbb{M}$ *is in* **L** *if and only if for all* $w_1, w_2, w_3 \in \mathbb{M}$, $w_3 w_2 w_1 = w_1$ *implies* $w_2 w_1 = w_1$.

Definition A.11. *The* ***syntactic congruence*** $\preceq_{\mathbb{L}}$ *is the equivalence relation on* Σ^* *given the language* $\mathbb{L}$ *such that for all* $\boldsymbol{x}, \boldsymbol{y} \in \Sigma^*$, *we have* $\boldsymbol{x} \preceq_{\mathbb{L}} \boldsymbol{y}$ *if and only if:*

$$\boldsymbol{sxz} \in \mathbb{L} \iff \boldsymbol{syz} \in \mathbb{L}\ \forall \boldsymbol{s}, \boldsymbol{z} \in \Sigma^* \tag{17}$$

Given a language, the syntactic congruence partitions strings of Σ^* into classes such that they are syntactically equivalent with respect to the language's automaton.

Example:
Consider the language PARITY (introduced in §3.4) and its syntactic congruence $\preceq_{\mathbb{L}}$. $\preceq_{\mathbb{L}}$ then has exactly two congruence classes, one for strings with an even numbers of 1s (which we denote by $\mathcal{C}_{\text{even}}$) and another for strings with an odd number of 1s (which we denote by $\mathcal{C}_{\text{odd}}$). Example strings in each class are:

$$011, 1111, 0101 \in \mathcal{C}_{\text{even}}$$
$$1, 10011, 1101 \in \mathcal{C}_{\text{odd}}$$

Definition A.12. *Let* $\mathbb{M}$ *be the free monoid,* $\mathbb{L}$ *be a regular language and* $\preceq_{\mathbb{L}}$ *its syntactic congruence. The* ***syntactic monoid*** *of* $\mathbb{L}$ *is the quotient monoid* $\mathbb{M}/\preceq_{\mathbb{L}}$.

Interestingly, the syntactic monoid of a regular language is isomorphic to the transition monoid of the minimal automaton accepting the language.

Finally, we now extend the monoid class **R** to account for modularity.

Definition A.13. *Let* $\mathbb{M}$ *be the syntactic monoid of a regular language* $\mathbb{L}$ *with associated automaton* $\mathcal{A}$. *Let* $\mathbb{M}_k$ *be the syntactic monoid of the language associated with* $\mathcal{A}_k$, *the k-automaton. Then,* $\mathbb{M}$ *is in* **QR** *if and only if there exists an integer* k *such that* $\mathbb{M}_k$ *is in* **R**.

A.6 Equivalences between Formalisms

To conclude this section, we summarize in Tab. 3 the known characterizations of $\mathbf{FO}^2[<, \mathtt{MOD}]$ and $\mathbf{PFO}^2[<]$, and introduce the analogous characterization of $\mathbf{PFO}^2[<, \mathtt{MOD}]$ which we prove in §B.

First-order logic	LTL	Regular expression	Syntactic Monoid	Automata	Note
$\mathbf{FO}^2[<, \mathtt{MOD}]$	$\mathbf{LTL}[\mathbf{P}, \mathbf{F}, \mathtt{MOD}]$	Unambiguous modular	$\mathbf{QDA}$	2-PODFA$_k$	Dartois & Paperman (2013)
$\mathbf{PFO}^2[<]$	$\mathbf{LTL}[\mathbf{P}]$	Left-deterministic	$\mathbf{R}$	PODFA	Brzozowski & Ellen (1980) and Li & Cotterell (2025)
$\mathbf{PFO}^2[<, \mathtt{MOD}]$	$\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$	Left-deterministic modular	$\mathbf{QR}$	PODFA$_k$	§B

Table 3: Characterizations of fragments of first-order logic. The equivalences in the last row are novel.

B Proofs: Characterizations of $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$

In this section, we provide a rich characterization of $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ via the following theorem.

Theorem 3.3. *Let $\mathbb{L} \subseteq \Sigma^*$ be a regular language, $\mathbb{M}$ be its syntactic monoid, and $\mathcal{A}$ be the minimal DFA accepting it. The following assertions are equivalent:*

(i) $\mathbb{L}$ *is a left-deterministic modular polynomial*
(ii) $\mathbb{M} \in \mathbf{QR}$
(iii) $\exists k \in \mathbb{N}$ *such that* $\mathcal{A}_k$ *is partially ordered*
(iv) $\mathbb{L} = \mathbb{L}(\phi)$ *for some* $\mathbf{PFO}^2[<, \mathtt{MOD}]$ *formula* ϕ
(v) $\mathbb{L} = \mathbb{L}(\psi)$ *for some* $\mathbf{LTL}[\mathbf{P}, \mathtt{MOD}]$ *formula* ψ

Most of the following lemmata are adaptations from equivalences proven in prior work (Dartois & Paperman, 2013; 2015; Li & Cotterell, 2025) applied on different fragments of first-order logic.

Lemma B.1. *A language $\mathbb{L}$ has its syntactic monoid in $\mathbf{QR}$ if and only if there exists a k such that $\mathbb{L}$'s automaton belongs to* PODFA$_k$.

Proof. By definition, a language $\mathbb{L}$ has its syntactic monoid in $\mathbf{QR}$ if and only if there exists a k such that its k-automaton $\mathcal{A}_k$ has its syntactic monoid in $\mathbf{R}$. The equivalence between the transition monoid of $\mathcal{A}_k$ being in $\mathbf{R}$ and $\mathcal{A}_k$ being a PODFA follows from Brzozowski & Ellen (1980). Therefore, $\mathbb{L}$'s automaton belongs to PODFA$_k$ for some k. ■

Lemma B.2. *If ϕ is a $\mathbf{PFO}^2[<, \mathtt{MOD}]$ formula, then the syntactic monoid of ϕ's language is in $\mathbf{QR}$. If a language's syntactic monoid is in $\mathbf{QR}$, it is the language of some $\mathbf{PFO}^2[<, \mathtt{MOD}]$ formula.*

Proof. The regular languages with a syntactic monoid in $\mathbf{R}$ are exactly the $\mathbf{PFO}^2[<]$ languages (Li & Cotterell, 2025). Moreover, for any logic characterized by a local variety $\mathbf{V}$, augmenting the logic with modular predicates yields a logic characterized by $\mathbf{QV}$ (Dartois & Paperman, 2015). Since $\mathbf{PFO}^2[<]$ is characterized by $\mathbf{R}$ and $\mathbf{R}$ is local (Almeida, 2025), augmenting $\mathbf{PFO}^2[<]$ with modular predicates gives the class characterized by $\mathbf{QR}$. ■

Lemma B.3. $\mathbf{PFO}^2[<, \texttt{MOD}]$ *and* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ *characterize the same class of languages.*

Proof. The equivalence without modular predicates has already been proven via structural induction (Li & Cotterell, 2025). Adding the same unary predicate MOD to both logics therefore preserves the equivalence. It remains to show that unary modular predicates in $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ can express 0-ary predicates in $\mathbf{PFO}^2[<, \texttt{MOD}]$.

Formulas in $\mathbf{LTL}[\mathbf{P}]$ are evaluated at the string position $N+1$. Moreover, N is congruent to r modulo m if and only if $N+1$ is congruent to $r+1$ modulo m. Therefore, evaluating in $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ the unary modular predicate $\texttt{MOD}_m^{r+1}$ at $N+1$ is equivalent to evaluating the 0-ary predicate $\texttt{MOD}_m^r$ in $\mathbf{PFO}^2[<, \texttt{MOD}]$. ∎

To prove the equivalence between $\mathbf{PFO}^2[<, \texttt{MOD}]$ and left-deterministic modular polynomials, we consider languages over *enriched alphabets* (Dartois & Paperman, 2013).

We denote by $\mathbf{PFO}^2[<, \texttt{MOD}_d]$ the restriction of $\mathbf{PFO}^2[<, \texttt{MOD}]$ with congruences modulo d. For any $\mathbf{PFO}^2[<, \texttt{MOD}]$ formula, there exists an integer d such that the formula is also $\mathbf{PFO}^2[<, \texttt{MOD}_d]$-definable—by for example setting d as the least common multiplier of all the moduli used.

Definition B.1 (Enriched alphabets; Dartois & Paperman 2013)**.** *The set* $\Sigma_d \stackrel{\text{def}}{=} \Sigma \times (\mathbb{Z}/d\mathbb{Z})$ *is the* ***enriched alphabet*** *of* Σ*, where* $\mathbb{Z}/d\mathbb{Z} \stackrel{\text{def}}{=} \{0, 1, \ldots, d-1\}$*. The projection* $\texttt{proj}\colon \Sigma_d^* \to \Sigma^*$ *removes the* $(\mathbb{Z}/d\mathbb{Z})$ *component of every symbol* $(a, i) \in \Sigma_d$ *of strings in* Σ_d^**.*

Definition B.2 (Well-formed words; Dartois & Paperman 2013)**.** *Given some alphabet* Σ*, the enriched word* $(w_0, i_0) \ldots (w_N, i_N)$ *is* ***well-formed*** *if* $i_j \equiv j \pmod{d}$ *for all* $0 \leq j \leq N$*. We denote by* $K \subset \Sigma_d^*$ *the set of well-formed words.*

Note that on K, the projection operator proj is a bijective application. Indeed, every word $\boldsymbol{w} = w_0 \cdots w_N \in \Sigma^*$ has a unique preimage in K: the word $(w_0,\ 0 \bmod d) \cdots (w_N,\ N \bmod d)$ obtained by annotating each position j with $j \bmod d$. Surjectivity is immediate, and injectivity holds because distinct well-formed words over Σ_d differ in at least one symbol and therefore project to distinct words.

We can now describe $\mathbf{PFO}^2[<, \texttt{MOD}]$ with enriched alphabets.[3]

Proposition B.1. $\mathbf{PFO}^2[<, \texttt{MOD}_d](\Sigma^*) = \texttt{proj}(\mathbf{PFO}^2[<](\Sigma_d^*) \cap K)$

Proof. The statement has been shown to be true for $\mathbf{FO}^2[<, \texttt{MOD}]$ (Dartois & Paperman, 2013). The result does not depend on how the binary predicate $<$ is used, and therefore it also holds for $\mathbf{PFO}^2[<, \texttt{MOD}]$. Modular predicates of the form $\texttt{MOD}_d^r$ can be computed by disjuncting over all atomic formulas of the form $\pi_{(w,r)}$ in $\mathbf{PFO}^2[<](\Sigma_d^*) \cap K$ for all $w \in \Sigma$. For the converse, formulas over exclusively well-formed words can be equivalently described by formulas over words in Σ^* with modular predicates. For instance, $\pi_{(w,r)}$ in $\mathbf{PFO}^2[<](\Sigma_d^*) \cap K$ can be written as $\pi_w \wedge \texttt{MOD}_d^r$ in $\mathbf{PFO}^2[<, \texttt{MOD}_d](\Sigma^*)$. ∎

We can now prove the following lemma.

Lemma B.4. *A language is a left-deterministic modular polynomial if and only if it can be described by a* $\mathbf{PFO}^2[<, \texttt{MOD}]$ *formula.*

Proof. Recall that for any $\mathbf{PFO}^2[<, \texttt{MOD}]$ formula describing a language $\mathbb{L}$, there exists a $\mathbf{PFO}^2[<, \texttt{MOD}_d]$ formula, for some integer d, that also describes $\mathbb{L}$. By Prop. B.1, $\mathbb{L}$ is therefore the projection of the well-formed words of a language $\mathbb{L}'$ definable by a $\mathbf{PFO}^2[<](\Sigma_d^*)$ formula. It remains to be shown via this projection that $\mathbb{L}$ can be equivalently described with a left-deterministic polynomial.

[3] We may write $\mathbf{PFO}^2[<](\Sigma^*)$ to specify that the logic is being used over the alphabet Σ.

Because $\mathbb{L}'$ is definable by a $\mathbf{PFO}^2[<](\Sigma_d^*)$ formula, it is a disjoint union of left-deterministic monomials over Σ_d (Li & Cotterell, 2025). Recall that over well-formed strings, `proj` is a bijection. Therefore, `proj` preserves disjoint unions. It thus suffices to show that `proj` maps each left-deterministic monomial over Σ_d to a left-deterministic *modular* monomial over Σ, and conversely.

Fix an enriched monomial $M = \Xi_0^* \, \overline{w}_1 \, \Xi_1^* \ldots \overline{w}_N \, \Xi_N^*$ over Σ_d, where $\Xi_i \subseteq \Sigma \times (\mathbb{Z}/d\mathbb{Z})$ and $\overline{w}_i = (w_i, r_i)$ for all $0 \leq i \leq N$, and set

$$\Sigma_{i,j} \stackrel{\text{def}}{=} \{w \mid (w, j) \in \Xi_i\}.$$

We claim that the projection of the well-formed words of M is exactly the modular monomial determined, in the sense of Def. A.3, by the markers w_i, the residues r_i, and the subsets $\Sigma_{i,j}$.

Forward ($\Rightarrow$). Let $\boldsymbol{w} = \texttt{proj}(\overline{\boldsymbol{w}})$ for a well-formed $\overline{\boldsymbol{w}} \in M$, factored as $\overline{\boldsymbol{w}} = \bar{u}_0 \, \overline{w_1} \, \bar{u}_1 \ldots \overline{w_N} \, \bar{u}_N$ with $\bar{u}_i \in \Xi_i^*$. Projecting yields $u = u_0 \, w_1 \, u_1 \ldots w_N \, u_N$. Well-formedness places $\overline{w_i} = (w_i, r_i)$ at a position $\equiv r_i \pmod{d}$, so each w_i stands at residue r_i in u. For any position p inside the block u_i, its enriched letter is $(w, p \bmod d) \in \Xi_i$, hence $w \in \Sigma_{i, p \bmod d}$. Thus u satisfies both conditions of Def. A.3 and lies in the modular monomial.

Reverse ($\Leftarrow$). Conversely, any u in that modular monomial admits a factorization meeting the same two conditions; tagging each position p of u with $p \bmod d$ produces the unique well-formed preimage $\bar{u}$ with $\texttt{proj}(\bar{u}) = u$, and those conditions place $\bar{u}$ in M. No rotation or partial-block bookkeeping arises, since Def. A.3 is stated over global positions and already absorbs each block's residue offset.

Finally, left-determinism transfers in both directions: for every $0 < i \leq N$,

$$\overline{w_i} \notin \Xi_{i-1} \iff (w_i, r_i) \notin \Xi_{i-1} \iff w_i \notin \Sigma_{i-1, r_i},$$

the right-hand side being exactly the condition of Def. A.4. Hence M is left-deterministic if and only if its projected modular monomial is left-deterministic, and `proj` restricts to a bijection between left-deterministic monomials over Σ_d and left-deterministic modular monomials over Σ. Applying this correspondence to each term of the disjoint union describing $\mathbb{L}'$, together with Prop. B.1, shows that $\mathbb{L}$ is in $\mathbf{PFO}^2[<, \texttt{MOD}]$ if and only if it is a left-deterministic modular polynomial. ■

B.1 Inexpressibility Results

Proposition B.2. *The locally testable language* $\Sigma^* ab \Sigma^*$ *with* $|\Sigma| > 2$ *is not in* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] = \mathbf{PFO}^2[<, \texttt{MOD}] = \mathbf{SMAT}[\text{RoPE}_{\text{P}}]$[4].

Proof. Consider the minimal DFA $\mathcal{A}$ for $\Sigma^* ab \Sigma^*$ with the alphabet $\Sigma = \{a, b, c\}$, illustrated in Fig. 6.

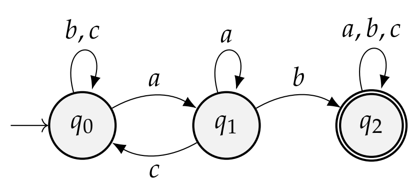

Figure 6: Minimal DFA for $\Sigma^* ab \Sigma^*$ over $\Sigma = \{a, b, c\}$

$\mathcal{A}_1 = \mathcal{A}$ can be seen to not be partially ordered. Consider some integer $k > 1$ and the k-automaton $\mathcal{A}_k$ of $\mathcal{A}$. Let $\boldsymbol{w}_a \in \Sigma^k$ be a string of k times the symbol a, and $\boldsymbol{w}_c \in \Sigma^k$ a string

[4] Note that over the two-letter alphabet $\Sigma = \{a, b\}$, $\Sigma^* ab \Sigma^*$ is in $\mathbf{PFO}^2[<]$ and thus also in $\mathbf{PFO}^2[<, \texttt{MOD}]$.

of k times the symbol c. We have that $\delta^k(q_0, \boldsymbol{w}_a) = q_1$ and $\delta^k(q_1, \boldsymbol{w}_c) = q_0$. Therefore, there is no partial order on the states of $\mathcal{A}_k$ as q_0 and q_1 are mutually reachable. Thus, for any integer k, $\mathcal{A}_k$ can never be partially ordered, and $\Sigma^* ab \Sigma^* \notin \mathbf{PFO}^2[<, \texttt{MOD}]$ for $|\Sigma| > 2$. ■

Proposition B.3. PARITY *is not in* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] = \mathbf{PFO}^2[<, \texttt{MOD}] = \textbf{SMAT[}\text{RoPE}_\text{P}\textbf{]}$.

Proof. Consider the minimal DFA $\mathcal{A}$ for PARITY, illustrated in Fig. 7.

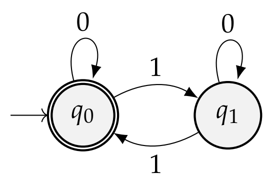

Figure 7: Minimal DFA for PARITY over $\Sigma = \{0, 1\}$

$\mathcal{A}_1 = \mathcal{A}$ can be seen to not be partially ordered. Consider some integer $k > 1$ and the k-automaton $\mathcal{A}_k$ of $\mathcal{A}$. Let $\boldsymbol{w}$ be a string of length k with an odd number of 1s. We have that $\delta^k(q_0, \boldsymbol{w}) = q_1$ and $\delta^k(q_1, \boldsymbol{w}) = q_0$. Therefore, there is no partial order on the states of $\mathcal{A}_k$ as q_0 and q_1 are mutually reachable. Thus, for any integer k, $\mathcal{A}_k$ can never be partially ordered, and PARITY$\notin \mathbf{PFO}^2[<, \texttt{MOD}]$. ■

C Proofs: Periodic RoPE

This section supplies the full proofs for periodic RoPE and for both inclusions summarized in §3: first $\textbf{SMAT[}\text{RoPE}_\text{P}\textbf{]} \subseteq \mathbf{PFO}^2[<, \texttt{MOD}]$, and then $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] \subseteq \textbf{SMAT[}\text{RoPE}_\text{P}\textbf{]}$. In all the following proofs, we follow the integer indexing convention for arrays: For $\boldsymbol{v} \in \mathbb{F}^D$ and a finite set of integers $\{i_1, \cdots i_n\}$, $\boldsymbol{v}[i_1, \cdots i_n] \in \mathbb{F}^n$ is the vector of elements in $\boldsymbol{v}$ indexed by $i_1, \cdots i_n$.

C.1 Component-Periodic Schedules

Proposition 2.1 (Component-periodicity via MOD)**.** *If a realized* RoPE *schedule* $\widehat{\Theta}$ *is component-periodic, then its image is finite and, for every matrix coordinate and realized value, the positions attaining that value are definable by unary modular predicates.*

Proof. Let $m_1, \ldots, m_{D/2}$ be component periods and $M \stackrel{\text{def}}{=} \operatorname{lcm}(m_1, \ldots, m_{D/2})$. Then $\widehat{\boldsymbol{R}}_{i+M} = \widehat{\boldsymbol{R}}_i$, so the image is contained in $\left\{\widehat{\boldsymbol{R}}_0, \ldots, \widehat{\boldsymbol{R}}_{M-1}\right\}$. For a matrix coordinate c and realized value v, define $A_{c,v} \stackrel{\text{def}}{=} \left\{a \in \{0, \ldots, M-1\} : \widehat{\boldsymbol{R}}_a[c] = v\right\}$. Then

$$\widehat{\boldsymbol{R}}_i[c] = v \quad \Longleftrightarrow \quad \bigvee_{a \in A_{c,v}} \texttt{MOD}^a_M(i)\,. \tag{18}$$

Thus every entry-value relation is definable with unary modular predicates. ■

Proposition 2.2 (Rational-angle functions are periodic)**.** *If* $\theta_d / (2\pi) = a_d / m_d \in \mathbb{Q}$*, then* $i \mapsto \mathbf{Rot}_{\theta_d, i}$ *has period* m_d*.*

Proof. We consider the mapping for cosine, as the sine analysis is the same. For some $i \in \mathbb{N}$, we have:

$$\cos(\theta_d i) = \cos\left(\frac{2\pi\theta_d}{2\pi} i\right) \tag{19a}$$
$$= \cos\left(2\pi \frac{a_d}{m_d} i\right) \tag{19b}$$
$$= \cos\left(2\pi \frac{a_d}{m_d} i + 2a_d\pi\right) \tag{19c}$$
$$= \cos\left(2\pi \frac{a_d}{m_d} (i + m_d)\right) \tag{19d}$$
$$= \cos(\theta_d(i + m_d)) \tag{19e}$$

Therefore, $\cos(\theta_d i)$ is periodic with integer period m_d. ■

C.2 Upper Bounding Periodic RoPE

Lemma 3.1. *Any* $\mathbb{F}$*-transformer with* RoPE$_\text{P}$ *can be simulated by* $\mathbf{PFO}^2[<, \texttt{MOD}]$. *Hence,* **SMAT[**RoPE$_\text{P}$**]** $\subseteq \mathbf{PFO}^2[<, \texttt{MOD}] = \mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$.

Proof. We formalize what it means for $\mathbf{PFO}^2[<, \texttt{MOD}]$ to *simulate* a transformer.

Definition C.1 (Li & Cotterell 2025, Definition B.1.). *A function* $\boldsymbol{\lambda}\colon (\overline{\Sigma})^* \to (\mathbb{F}^D)^*$ *is said to be* ***simulated*** *by* $\mathbf{PFO}^2[<, \texttt{MOD}]$ *if for every dimension* $d \in [D]$ *and every floating-point value* $f \in \mathbb{F}$*, there exists a single-variable formula* ϕ *in* $\mathbf{PFO}^2[<, \texttt{MOD}]$ *such that for every position* $i \in [N]$*:*

$$\boldsymbol{\lambda}(\boldsymbol{w})_{d,i} = f \iff \boldsymbol{w}, i \models \phi \tag{20}$$

Similarly, we can define the simulation of a matrix-valued function via a two-variable $\mathbf{PFO}^2[<, \texttt{MOD}]$ *formula.*

We prove that $\mathbf{PFO}^2[<, \texttt{MOD}]$ can simulate any RoPE$_\text{P}$ transformer. First, each realized matrix entry can be simulated by the formulas from Prop. 2.1.

A realized rotation matrix $\widehat{\boldsymbol{R}}_i$ can be seen as a function mapping positions i to elements in $\mathbb{F}^{D\times D}$. For every matrix coordinate $c \in [D \times D]$ and floating-point value $f \in \mathbb{F}$, we define a $\mathbf{PFO}^2[<, \texttt{MOD}]$ formula $\phi_{c,f}$ such that:

$$\widehat{\boldsymbol{R}}_i[c] = f \iff \boldsymbol{w}, i \models \phi_{c,f} \tag{21}$$

0-elements of the matrix: For the 0-elements of $\widehat{\boldsymbol{R}}_i$, we create constant formulas $\phi_\top \stackrel{\text{def}}{=} \top$ and $\phi_\bot \stackrel{\text{def}}{=} \bot$. If the floating-point value f is 0, we assign the constant formula $\phi_\top$. Otherwise, we assign $\phi_\bot$.

Position-dependent elements of the matrix. For every coordinate c and realized value v, the formula $\bigvee_{a \in A_{c,v}} \texttt{MOD}_M^a(i)$ is true exactly where $\widehat{\boldsymbol{R}}_i[c] = v$ (Prop. 2.1). This covers every block entry without recovering an ideal angle from its finite-precision realization.

Therefore, we can simulate the rotation matrices with $\mathbf{PFO}^2[<, \texttt{MOD}]$. Omitting the periodic rotations, all other components of a fixed precision soft-attention transformer can be simulated by $\mathbf{PFO}^2[<]$ (Li & Cotterell, 2025). Finally, as the elements of the rotation matrices belong to a finite set (due to Prop. 2.1), the rotation matrices belong themselves to some finite set. Therefore, the rotation of the keys and queries is a mapping between two finite sets. Any mapping between finite sets can be simulated via a disjunction over the finitely-many inputs in the domain that lead to some output in the image (Chiang et al., 2023; Li & Cotterell, 2025). This concludes the proof. ■

Proposition 3.1. *Binary modular predicates can be expressed with unary modular predicates and Boolean logic.*

Proof. We will show that $\texttt{MOD}_m^r(i,j)$ can be written as the following finite disjunction.

$$\texttt{MOD}_m^r(i,j) = \bigvee_{a=0}^{m-1} \left(\texttt{MOD}_m^a(i) \wedge \texttt{MOD}_m^{(a+r) \bmod m}(j)\right) \tag{22}$$

We denote by i_0 and j_0 the residues such that $i \equiv i_0 \pmod m$ and $j \equiv j_0 \pmod m$.

$\texttt{MOD}_m^r(i,j) = \top$ if and only if $(j-i) \equiv r \pmod m$. Equivalently, $(j_0 - i_0) \equiv r \pmod m$, i.e. $j_0 \equiv r + i_0 \pmod m$. Therefore, $\texttt{MOD}_m^r(i,j) = \top$ if and only if there exist residues i_0 and j_0 such that 1) $i \equiv i_0 \pmod m$ 2) $j \equiv j_0 \pmod m$ with $j_0 \equiv i_0 + r \pmod m$. This statement is exactly the right hand side of the given equation. ■

C.3 Lower Bounding Periodic RoPE

On finite precision. Before proving the lower bound $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] \subseteq \mathbf{SMAT}[\mathrm{RoPE}_\mathrm{P}]$, we recall that our result holds for *some* finite-precision regime $\mathbb{F}$. In other words, we design a tailored finite set of values $\mathbb{F}$ that works for our constructions. This assumption is implicitly made in related work (Yang et al., 2024). This allows us to assume that the finitely many values our constructions manipulate are exactly representable, i.e. are present in $\mathbb{F}$. For each modulus m, we specify the required roots of unity directly as an m-entry rotation table and include its finitely many entries in $\mathbb{F}$. A cyclic counter or modular index return the corresponding table entry. This establishes an exact result in the abstract expressivity model; it does not assert that standard floating-point hardware represents $2\pi/m$, the corresponding roots of unity, or the unbounded product $2\pi i/m$ exactly. The experiments in §3.5 test a practical finite-precision implementation empirically.

Lemma 3.2. *For every* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ *formula, there is an* $\mathbb{F}$*-transformer with* RoPE_P *that simulates the formula. Hence* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] \subseteq \mathbf{SMAT}[\mathrm{RoPE}_\mathrm{P}]$.

Proof. We formalize what it means for $\mathbf{SMAT}[\mathrm{RoPE}_\mathrm{P}]$ to *simulate* a $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ formula.

Definition C.2 (Li & Cotterell 2025, Definition B.9.)**.** *A* $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ *formula* ψ *is simulated by a function* $\boldsymbol{\lambda}\colon (\overline{\Sigma})^* \to (\mathbb{F}^D)^*$ *if there exists a coordinate* $d \in [D]$ *such that for every input string* $\boldsymbol{w} \in (\overline{\Sigma})^*$ *and string position* $i \in [N]$*:*

$$\boldsymbol{\lambda}(\boldsymbol{w})_{d,i} = \begin{cases} 1 \text{ if } \boldsymbol{w}, i \models \psi \\ 0 \text{ otherwise} \end{cases} \tag{23}$$

We proceed by structural induction on the $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ formula.

Induction on $\mathbf{LTL}[\mathbf{P}]$ without $\texttt{MOD}$. Any $\mathbf{LTL}[\mathbf{P}]$ formula can be simulated by a transformer (without PEs) via structural induction on its constituents (Li & Cotterell, 2025). To lift this to $\mathbf{SMAT}[\mathrm{RoPE}_\mathrm{P}]$, we allocate dedicated dimensions for the Boolean connectives and **P**—the sole temporal operator of $\mathbf{LTL}[\mathbf{P}]$—and set their rotation angles to 0 (identity matrices), so that RoPE has no effect on those dimensions. We then apply Li & Cotterell's (2025) simulation directly, yielding $\mathbf{LTL}[\mathbf{P}] \subseteq \mathbf{SMAT}[\mathrm{RoPE}_\mathrm{P}]$.

Simulating modular predicates. It remains to realize each modular atom. Consider the formula $\texttt{MOD}_m^r(i)$. We isolate the case $i = 1$, where $\texttt{MOD}_m^r(1)$ is then a *compile-time constant*. The first position can be flagged by performing uniform attention over the strict left context and checking the count is exactly 1, which we assume is in $\mathbb{F}$. We can then store $\texttt{MOD}_m^r(1)$ at that position. Moreover, the case $m = 1$ is the constant-true predicate, so we proceed to the $m \geq 2, i \geq 2$ case.

We first allocate one rotary pair at dimensions $d, d+1$ with the exact period-m lookup table (see §2.4)

$$
\begin{aligned}
T_m(s) &\stackrel{\text{def}}{=} \mathbf{Rot}_{2\pi/m,s} \quad (0 \le s < m), &\qquad \widehat{\boldsymbol{R}}_i^{(d)} &\stackrel{\text{def}}{=} T_m(i \bmod m), \\
\delta_m &\stackrel{\text{def}}{=} \min_{1 \le s < m} (1 - \cos(2\pi s/m)) > 0, &\qquad b_m &\stackrel{\text{def}}{=} 1 - \delta_m/2.
\end{aligned}
\tag{24}
$$

We also allocate one unrotated pair of dimensions $d', d'+1$. All other dimensions are unused (i.e., manually set to 0 in keys and queries) in our construction.

<u>*Query*</u>. Before the rotations, we set the elements of the query projection to position-independent constants. These constants are implemented by setting the query weight matrix $\boldsymbol{Q}$ to a zero-matrix and encoding the desired values in the query bias vector $\boldsymbol{b}_q$, so that $\boldsymbol{Q}\boldsymbol{x}_i^\ell + \boldsymbol{b}_q = \boldsymbol{b}_q$ for all inputs $\boldsymbol{x}_i^\ell$.

$$
\begin{aligned}
(\boldsymbol{Q}\boldsymbol{x}_i^\ell)[d, d+1] &= \begin{pmatrix} \cos(2\pi r/m) \\ -\sin(2\pi r/m) \end{pmatrix} \\
(\boldsymbol{Q}\boldsymbol{x}_i^\ell)[d', d'+1] &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}
\end{aligned}
$$

The post-rotation query dimensions then become:

$$
\begin{aligned}
\boldsymbol{q}_i^\ell[d, d+1] &= \begin{pmatrix} \cos\left(\frac{2\pi i}{m}\right)\cos\left(\frac{2\pi r}{m}\right) + \sin\left(\frac{2\pi i}{m}\right)\sin\left(\frac{2\pi r}{m}\right) \\ \sin\left(\frac{2\pi i}{m}\right)\cos\left(\frac{2\pi r}{m}\right) - \cos\left(\frac{2\pi i}{m}\right)\sin\left(\frac{2\pi r}{m}\right) \end{pmatrix} \\
\boldsymbol{q}_i^\ell[d', d'+1] &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}
\end{aligned}
$$

<u>*Key.*</u> We set the pre-rotation dimensions of the key vector as follows.

$$
\begin{aligned}
(\boldsymbol{K}\boldsymbol{x}_j^\ell)[d, d+1] &= \begin{pmatrix} C\mathbb{1}\{w_j = \textsc{bos}\} \\ 0 \end{pmatrix} \\
(\boldsymbol{K}\boldsymbol{x}_j^\ell)[d', d'+1] &= \begin{pmatrix} Cb_m(1 - \mathbb{1}\{w_j = \textsc{bos}\}) \\ 0 \end{pmatrix}
\end{aligned}
$$

Where C is a positive constant in $\mathbb{F}$ defined later, and $\mathbb{1}\{w_j = \textsc{bos}\}$ can be computed via a feedforward network.

Note that because BOS correspond to position 0, the dimensions $d, d+1$ are unaffected by rotation: At BOS the rotation is an identity mapping, at a symbol position the vector is a 0-vector. Moreover, the dimensions $d', d'+1$ are by construction unrotated.

<u>*Score.*</u> The query–key computation score yields:

$$
(\boldsymbol{q}_i^\ell)^\intercal \boldsymbol{k}_j^\ell = \begin{cases} C\cos(2\pi(i-r)/m), & \text{if } w_j = \textsc{bos} \\ Cb_m, & \text{otherwise} \end{cases}
\tag{25}
$$

Notice that $\cos(2\pi(i-r)/m)$ is maximized when $i \equiv r \pmod m$.

<u>*Modular predicate condition satisfied.*</u> If $i \equiv r \pmod m$, the score with BOS is C and the scores with non-BOS positions are all $C(1 - \delta_m/2)$. Therefore, the largest score is C by a margin of $C\delta_m/2$.

<u>*Modular predicate condition not satisfied.*</u> Suppose $i \not\equiv r \pmod m$. Then $\cos(2\pi(i-r)/m) \le 1 - \delta_m$ by definition of δ_m, so the score with BOS satisfies $C\cos(2\pi(i-r)/m) \le C(1-\delta_m)$. Every non-BOS position still scores exactly $Cb_m = C(1-\delta_m/2)$, independently of j. Hence every non-BOS position now beats BOS, by a margin of at least

$$
C(1-\delta_m/2) - C(1-\delta_m) = C\delta_m/2.
\tag{26}
$$

So, in either case ($i \equiv r \pmod m$ or not), whichever side wins—BOS or the (tied) non-BOS positions—does so by a margin of at least $C\delta_m/2$.

Choosing C and extracting the predicate. We define the value vector to be 1 at BOS and 0 at every other position, so that the attention output at position i is exactly $\alpha_{i,\text{BOS}}$. Selecting $C \geq 2f_{\text{large}}/\delta_m$ makes the margin $C\delta_m/2$ established above at least f_{large} in both cases. By stable softmax (§2), subtracting the maximum score from every score leaves an exponent of 0 at every position attaining the maximum and an exponent at most $-f_{\text{large}}$—hence rounding to $\exp(-f_{\text{large}}) = 0$—at every position on the losing side of the margin.

- If $i \equiv r \pmod{m}$: BOS uniquely attains the maximum score, so $\alpha_{i,\text{BOS}} = 1$ and $\alpha_{i,j} = 0$ for every non-BOS j. The attention output is 1.
- If $i \not\equiv r \pmod{m}$: Every non-BOS position attains the (tied) maximum score, so $\alpha_{i,\text{BOS}} = 0$, while the non-BOS weights $\alpha_{i,j}$ absorb the remaining mass. The attention output is still $\alpha_{i,\text{BOS}} = 0$, regardless of how stable softmax splits weight among the tied non-BOS positions.

The attention output then equals $\mathbb{1}\{\texttt{MOD}^r_m(i)\}$: The rotary pair $d, d+1$ supplies the position-dependent score against BOS, while the unrotated pair $d', d'+1$ supplies the constant non-BOS baseline Cb_m that this score is compared against, and together they correctly compute $\texttt{MOD}^r_m(i)$.

Completing the induction. A formula contains only finitely many modular atoms, so it requires finitely many rotary pairs and constants. Choose $\mathbb{F}$ to contain the entries of their finite lookup tables and all displayed intermediate values. This completes the induction. ■

C.4 Characterizing Periodic SiPE

Theorem 3.2. $\textbf{SMAT}[\text{SiPE}_\text{P}] = \textbf{LTL}[\mathbf{P}, \texttt{MOD}] = \textbf{PFO}^2[<, \texttt{MOD}]$.

Proof. We now show both directions of the equivalence.

Forward ($\Rightarrow$). Every non-PE component of a **SMAT[**SiPE$_\text{P}$**]** transformer is in $\textbf{PFO}^2[<]$ (Li & Cotterell, 2025). Periodic sinusoidal position encodings can also be expressed via unary modular predicates by Prop. 2.1, yielding the inclusion in $\textbf{PFO}^2[<, \texttt{MOD}]$.

Reverse ($\Leftarrow$). NoPE transformers, which are subsumed by **SMAT[**SiPE$_\text{P}$**]** transformers, can express $\textbf{PFO}^2[<]$. Simulating modular predicates can be done by processing sinusoidal position encodings with feedforward networks (Yang et al., 2024) using an appropriate schedule and pre-computed lookup table (§2.4), leading to the reverse direction. Note that Yang et al.'s (2024) modular predicate simulation does not rely on the attention type. ■

D Proofs: Non-Periodic RoPE

Because Cor. 4.1 certifies the simulation only up to a finite length, we first record the length-bounded variant of Def. C.2 that it uses.

Definition D.1 (Bounded-length simulation)**.** *Let $L \in \mathbb{N}$. A formula ψ is L-**simulated** by a function $\boldsymbol{\lambda}\colon (\overline{\Sigma})^* \to (\mathbb{F}^D)^*$ if there exists a coordinate $d \in [D]$ such that for every input BOS$\boldsymbol{w}$EOS with $|\boldsymbol{w}| \leq L$ and every position $i \in [|\boldsymbol{w}|+1]$,*

$$\boldsymbol{\lambda}\left(\text{BOS}\boldsymbol{w}\text{EOS}\right)_{d,i} = \begin{cases} 1 & \textit{if } \boldsymbol{w}, i \models \psi \\ 0 & \textit{otherwise.} \end{cases} \tag{27}$$

Taking $L \to \infty$ recovers Def. C.2.

Throughout, positions are indexed so that BOS occupies position 0, the string occupies positions $1, \ldots, |\boldsymbol{w}|$, and EOS occupies position $|\boldsymbol{w}|+1$, which is the readout position. Hence $|\boldsymbol{w}| \leq L$ makes every query position at most $L+1$, which is exactly the range quantified over in Eq. (10).

Corollary 4.1. *Fix g, C, and $\mathbb{F}$, and let $L \in \mathbb{N}$. Write* **SMAT**[RoPE$_{\mathrm{NP}}$]*-transformers for soft-attention* RoPE$_{\mathrm{NP}}$ *transformers with unrotated subspaces, and let L-simulation be as in Def. D.1.*

(i) For every offset $k \geq 1$ with $k \leq L \leq N^{(k)}_{\max}(g, C, \mathbb{F})$, such a transformer L-simulates the fixed-offset operator $\boldsymbol{Y}^k$.

(ii) If $1 \leq L \leq N^{(1)}_{\max}(g, C, \mathbb{F})$, it L-simulates every **LTL**[**P**, $\boldsymbol{Y}$]*-definable formula. More generally, for a finite set of offsets $K \subseteq \mathbb{N}_{\geq 1}$ with $\max K \leq L \leq \min_{k \in K} N^{(k)}_{\max}(g, C, \mathbb{F})$, it L-simulates every formula built from atomic predicates, Boolean connectives,* **P**, *and the operators $\{\boldsymbol{Y}^k : k \in K\}$.*

(iii) Moreover, for every $L \in \mathbb{N}$ and every finite offset set K there exist a rotation angle g with $g/(2\pi) \notin \mathbb{Q}$, scales C, and a finite-precision regime $\mathbb{F}$ satisfying the hypothesis of (ii).

Proof. Allocate unrotated dimensions to the standard structural-induction simulation of **LTL**[**P**] (Li & Cotterell, 2025). Fix $k \geq 1$. To simulate a $\mathbf{Y}^k\psi$ subformula directly, use one rotary pair $d, d+1$ in a dedicated attention head, with query bias $C(\cos(kg), -\sin(kg))$ and key bias $(1, 0)$ at those two coordinates. In this head we set $\boldsymbol{Q}$ and $\boldsymbol{K}$ to the zero matrix and set every coordinate of $\boldsymbol{b}_q$ and $\boldsymbol{b}_k$ outside the pair $d, d+1$ to 0. The score is therefore exactly the contribution of that pair: The unrotated dimensions carrying the **LTL**[**P**] simulation, and every other rotary pair, contribute 0 to it. In particular the biases are the head's only source of query and key content, so the score is position-dependent but content-independent, as the fixed-offset semantics of $\mathbf{Y}^k$ requires.

After rotation at query position i and key position j, exact arithmetic gives

$$\boldsymbol{S}^{g,C,k}_{i,j} = C\cos\left(g\left(i - k - j\right)\right), \tag{28}$$

which is Eq. (9). Under the model's actual arithmetic, denote the realized score by $\widehat{\boldsymbol{S}}^{g,C,k,\mathbb{F}}_{i,j}$ as in Eq. (10). Store the truth value of ψ in the value vector at each string position and store 0 at BOS.

We prove (i). Fix $k \leq L \leq N^{(k)}_{\max}(g, C, \mathbb{F})$. At every $k + 1 \leq i \leq L + 1$, position $i - k$ beats every other key by at least f_{large}, by the definition of $N^{(k)}_{\max}$; note that the competing keys include BOS at $j = 0$, whose rotation matrix is the identity, so its key is $(\cos 0, \sin 0) = (1, 0)$. Stable softmax therefore assigns weight 1 to $i - k$ and 0 elsewhere, so the head returns the truth value of ψ exactly k positions earlier. Since $i \geq k + 1$, the attended position $i - k$ is a string position, not BOS. At the boundary $i \leq k$, the semantics require $\mathbf{Y}^k\psi$ to be false. The unrotated construction simulates the gate $\mathbf{P}^k\top$, which holds exactly when $i > k$—indeed $\mathbf{P}\top$ holds at i iff $i > 1$, and inductively $\mathbf{P}^k\top = \mathbf{P}\left(\mathbf{P}^{k-1}\top\right)$ holds at i iff some $j < i$ has $j > k - 1$, i.e. iff $i > k$; a feedforward Boolean conjunction with this gate therefore suppresses the arbitrary attention output at those boundary positions, mapping it to 0 regardless of its value. This L-simulates $\mathbf{Y}^k$, proving (i).

For (ii), every **LTL**[**P**, **Y**] formula is built from atomic predicates, Boolean connectives, **P**, and **Y**, so it needs only the offset $k = 1$: one layer per **Y** occurrence, applying (i) with $k = 1$, which is available because $L \leq N^{(1)}_{\max}(g, C, \mathbb{F})$ by hypothesis. Together with the unrotated **P** and Boolean constructions this completes the structural induction. For a finite offset set K, give each $k \in K$ its own head and rotary pair; the hypothesis $L \leq \min_{k \in K} N^{(k)}_{\max}(g, C, \mathbb{F})$ makes (i) available for every $k \in K$ simultaneously, and $\max K \leq L$ makes each application well posed.

For (iii), fix any finite L and finite collection of offsets, and choose $g/(2\pi) \notin \mathbb{Q}$. Then $\delta_L(g) > 0$. Choose the scales C and a finite-precision regime jointly so that $C\delta_L(g) \geq f_{\mathrm{large}}$ for every corresponding head and all parameters and intermediate sine, cosine, product, and score values used by the constructions through positions $L + 1$ are represented exactly. This is possible in the abstract finite-value model because only finitely many offsets and nonzero values are required, while the regime can retain a separate underflow cell around zero. On that finite range the ideal identities are evaluated exactly, so Eq. (10) gives $N_{\max}^{(k)}(g, C, \mathbb{F}) \geq L$ for every selected k. ■

E Experiments

We supplement the periodic and conventional non-periodic experiments in §§ 3.5 and 4.4 with additional details.

E.1 Languages

We first introduce the different languages we train our transformers on.

- $(ab)^*$: the language of strings in which the substring ab is repeated arbitrarily many times. It is star-free, inside $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}] = \mathbf{PFO}^2[<, \texttt{MOD}]$ and outside of $\mathbf{PFO}^2[<]$. It is also the Dyck language (well-balanced parentheses) with 1 pair of parentheses, limited to depth 1.
- $(aa)^*$: the language of strings of even length. It is a canonical example of a non-star-free, $\mathbf{LTL}[\mathbf{P}, \texttt{MOD}]$ language.
- $a\Sigma^*$: the language of strings starting with a designated symbol a over some alphabet Σ such that $|\Sigma| > 1$. It is the simplest language of the class $\mathbf{PFO}^2[<] = \mathbf{LTL}[\mathbf{P}]$.
- Σ^*a: the language of strings ending with a designated symbol a over some alphabet Σ such that $|\Sigma| > 1$. It belongs in $\mathbf{LTL}[\mathbf{F}]$ and $\mathbf{LTL}[\mathbf{Y}]$.
- $\Sigma^*ab\Sigma^*$: the language of strings containing the contiguous substring ab. It is locally testable and definable in $\mathbf{LTL}[\mathbf{P}, \mathbf{Y}]$.

E.2 Experimental Setup

Our experimental setup follows Delétang et al. (2023); Li & Cotterell (2025). Our transformers use soft attention and strict future masking, with 5 layers, model dimension set to 64, and 8 attention heads. Training strings are of length up to 40, while test strings range from length 41 to 500. The model is trained for 1,000,000 steps with a batch size of 128. For evaluation, we generate 512 samples per test length. Each experiment is run with 5 different random seeds and 3 learning rates ($1 \times 10^{-4}, 3 \times 10^{-4}, 5 \times 10^{-4}$).

E.3 Periodic-Construction Settings

The periodic experiments in Tab. 2 compare the period-targeting RoPE_P and SiPE_P variants with NoPE; the conventional fully rotated controls appear in Fig. 2. The component-periodic scheduler is explicitly written out in the code implementation.

E.4 Conventional Non-Periodic Settings

Recall in RoPE's initial presentation that the base value is set to 10^4. Prior extrapolation experiments find that this standard choice can be the worst tested base, with improvements on both sides of 10^4 (Liu et al., 2024). To test whether our conclusions depend on this choice, we therefore sweep bases $\{10^{-12}, 10^{-8}, 10^{-4}, 1, 10^4, 10^8, 10^{12}\}$. Fig. 3 reports every fully rotated RoPE_NP configuration and compares it with NoPE. Fig. 2 reports fully rotated RoPE_NP and SiPE_NP configurations across the same sweep for $(aa)^*$ and $(ab)^*$.

E.5 Detailed Results

Tabs. 4 and 5 report the complete aggregate statistics for the tasks and fully rotated settings shown in Figs. 2 and 3. Each row aggregates 5 seeds and 3 learning rates.

Table 4: Detailed results for the fully rotated conventional base sweep on the modular languages in Fig. 2.

Language	PE	Base	Accuracy			N^*		
			Max	Mean	Std.	Max	Mean	Std.
$(aa)^*$	RoPE_{NP}	10^{-12}	0.522	0.489	0.0204	53.0	41.3	11.6
$(aa)^*$	RoPE_{NP}	10^{-8}	0.500	0.483	0.0069	65.0	50.1	14.9
$(aa)^*$	RoPE_{NP}	10^{-4}	0.513	0.501	0.0061	57.0	49.5	3.1
$(aa)^*$	RoPE_{NP}	1	0.509	0.494	0.0076	47.0	34.3	17.2
$(aa)^*$	RoPE_{NP}	10^{4}	0.507	0.489	0.0117	48.0	26.3	21.5
$(aa)^*$	RoPE_{NP}	10^{8}	0.515	0.490	0.0146	46.0	22.7	21.3
$(aa)^*$	RoPE_{NP}	10^{12}	0.517	0.494	0.0095	49.0	23.3	21.9
$(aa)^*$	SiPE_{NP}	10^{-12}	0.520	0.504	0.0117	41.0	16.4	20.1
$(aa)^*$	SiPE_{NP}	10^{-8}	0.654	0.619	0.0218	88.0	71.3	11.9
$(aa)^*$	SiPE_{NP}	10^{-4}	**0.657**	**0.623**	0.0273	**90.0**	**75.7**	6.3
$(aa)^*$	SiPE_{NP}	1	0.509	0.496	0.0089	43.0	8.5	16.9
$(aa)^*$	SiPE_{NP}	10^{4}	0.522	0.503	0.0078	41.0	2.7	10.2
$(aa)^*$	SiPE_{NP}	10^{8}	0.517	0.500	0.0049	0.0	0.0	0.0
$(aa)^*$	SiPE_{NP}	10^{12}	0.515	0.497	0.0107	0.0	0.0	0.0
$(ab)^*$	RoPE_{NP}	10^{-12}	0.572	0.530	0.0168	**98.0**	65.1	13.0
$(ab)^*$	RoPE_{NP}	10^{-8}	0.520	0.510	0.0057	58.0	48.9	5.2
$(ab)^*$	RoPE_{NP}	10^{-4}	0.530	0.514	0.0060	68.0	52.8	5.6
$(ab)^*$	RoPE_{NP}	1	0.522	0.509	0.0041	56.0	48.1	3.1
$(ab)^*$	RoPE_{NP}	10^{4}	0.740	0.560	0.0602	96.0	**66.0**	15.7
$(ab)^*$	RoPE_{NP}	10^{8}	**0.817**	**0.566**	0.0896	**98.0**	57.1	14.7
$(ab)^*$	RoPE_{NP}	10^{12}	0.703	0.524	0.0484	72.0	50.1	6.5
$(ab)^*$	SiPE_{NP}	10^{-12}	0.515	0.504	0.0050	42.0	16.8	20.6
$(ab)^*$	SiPE_{NP}	10^{-8}	0.648	0.536	0.0416	54.0	41.9	16.7
$(ab)^*$	SiPE_{NP}	10^{-4}	0.547	0.523	0.0145	70.0	52.9	7.6
$(ab)^*$	SiPE_{NP}	1	0.515	0.505	0.0045	48.0	23.1	21.6
$(ab)^*$	SiPE_{NP}	10^{4}	0.515	0.503	0.0044	46.0	17.1	20.9
$(ab)^*$	SiPE_{NP}	10^{8}	0.559	0.514	0.0142	52.0	26.3	21.6
$(ab)^*$	SiPE_{NP}	10^{12}	0.586	0.514	0.0204	52.0	24.1	22.8

Table 5: Detailed results for NoPE and the fully rotated conventional RoPE base sweep in Fig. 3.

Language	PE	Base	Accuracy			N^*		
			Max	Mean	Std.	Max	Mean	Std.
$\Sigma^* a$	NoPE	–	0.604	0.564	0.0237	60.0	50.8	3.9
$\Sigma^* a$	RoPE_{NP}	10^{-12}	0.741	0.709	0.0181	149.0	95.0	26.6
$\Sigma^* a$	RoPE_{NP}	10^{-8}	**0.927**	**0.829**	0.0898	116.0	81.3	16.8
$\Sigma^* a$	RoPE_{NP}	10^{-4}	0.913	0.785	0.0538	**224.0**	**99.8**	46.8
$\Sigma^* a$	RoPE_{NP}	1	0.753	0.649	0.0483	82.0	54.8	9.2
$\Sigma^* a$	RoPE_{NP}	10^{4}	0.555	0.524	0.0152	51.0	45.5	2.3
$\Sigma^* a$	RoPE_{NP}	10^{8}	0.890	0.733	0.1003	112.0	65.7	14.6
$\Sigma^* a$	RoPE_{NP}	10^{12}	0.685	0.640	0.0305	70.0	58.9	4.8
$a\Sigma^*$	NoPE	–	**1.000**	**0.989**	0.0413	**500.0**	**470.5**	82.0
$a\Sigma^*$	RoPE_{NP}	10^{-12}	**1.000**	0.977	0.0253	**500.0**	199.5	99.1
$a\Sigma^*$	RoPE_{NP}	10^{-8}	**1.000**	0.927	0.1033	**500.0**	166.7	101.4
$a\Sigma^*$	RoPE_{NP}	10^{-4}	**1.000**	0.961	0.0515	**500.0**	199.1	95.4
$a\Sigma^*$	RoPE_{NP}	1	**1.000**	0.953	0.1005	**500.0**	204.9	109.5
$a\Sigma^*$	RoPE_{NP}	10^{4}	**1.000**	0.847	0.1449	184.0	146.9	30.2

Continued on next page

Table 5 continued

Language	PE	Base	Accuracy			N^*		
			Max	Mean	Std.	Max	Mean	Std.
$a\Sigma^*$	RoPE_{NP}	10^8	**1.000**	0.928	0.0864	**500.0**	293.3	121.4
$a\Sigma^*$	RoPE_{NP}	10^{12}	**1.000**	0.988	0.0254	**500.0**	402.9	125.1
$\Sigma^*ab\Sigma^*$	NoPE	–	**0.874**	**0.672**	0.0829	**200.0**	**92.9**	32.6
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^{-12}	0.770	0.625	0.0697	103.0	61.9	16.8
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^{-8}	0.753	0.604	0.0812	76.0	58.7	9.8
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^{-4}	0.740	0.606	0.0556	73.0	55.2	7.4
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	1	0.647	0.624	0.0221	51.0	46.6	1.5
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^4	0.612	0.572	0.0250	53.0	49.8	3.5
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^8	0.624	0.594	0.0183	53.0	47.2	2.6
$\Sigma^*ab\Sigma^*$	RoPE_{NP}	10^{12}	0.658	0.637	0.0102	50.0	46.1	2.2